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Model-Based Analysis: Foundations of Correlation & Regression Modeling

Goals

- Develop basic concepts of linear regression from a probabilistic framework
- Estimating parameters and hypothesis testing with linear models

Regression

- Technique used for the modeling and analysis of numerical data

- Exploits the relationship between two or more variables so that we can gain information about one of them through knowing values of the other

- Regression can be used for prediction, estimation, hypothesis testing, and modeling causal relationships

Regression Lingo

$$Y = X_1 + X_2 + X_3$$

Dependent Variable	Independent Variable
Outcome Variable	Predictor Variable
Response Variable	Explanatory Variable

Why Linear Regression?

- Suppose we want to model the dependent variable Y in terms of three predictors, X_1, X_2, X

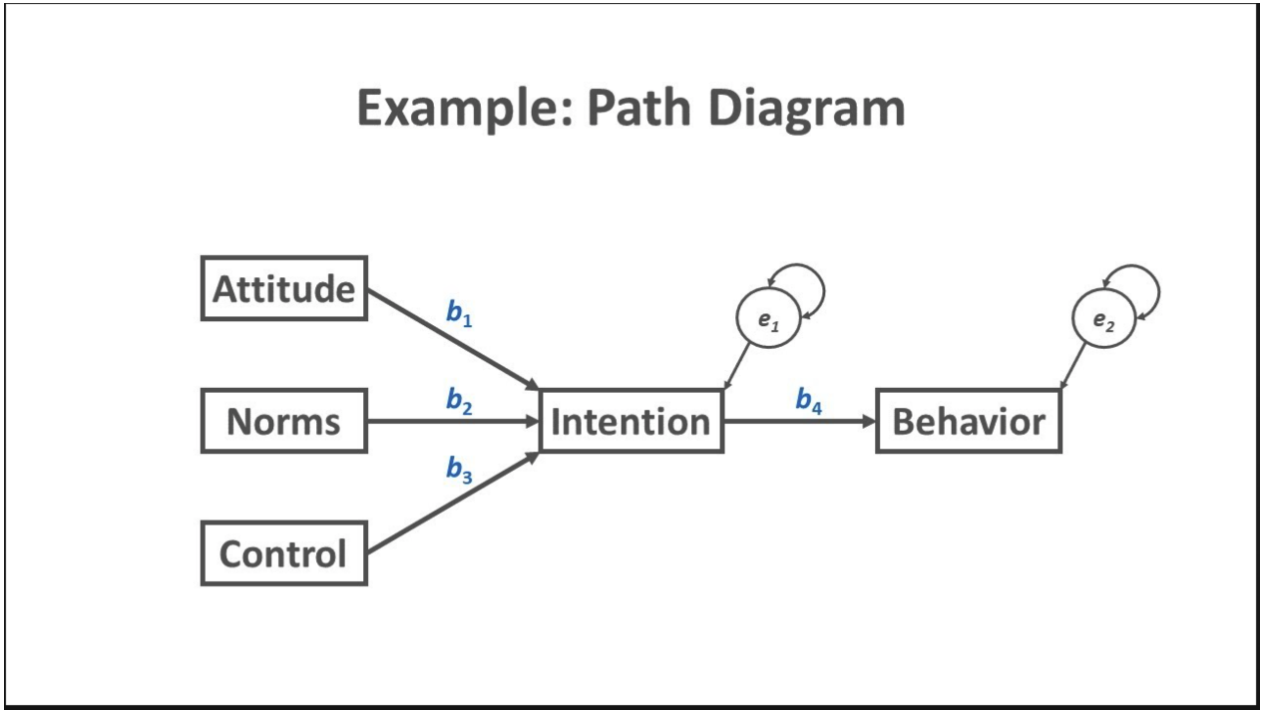
$$Y = f(X_1, X_2, X_3)$$

- Typically will not have enough data to try and directly estimate f

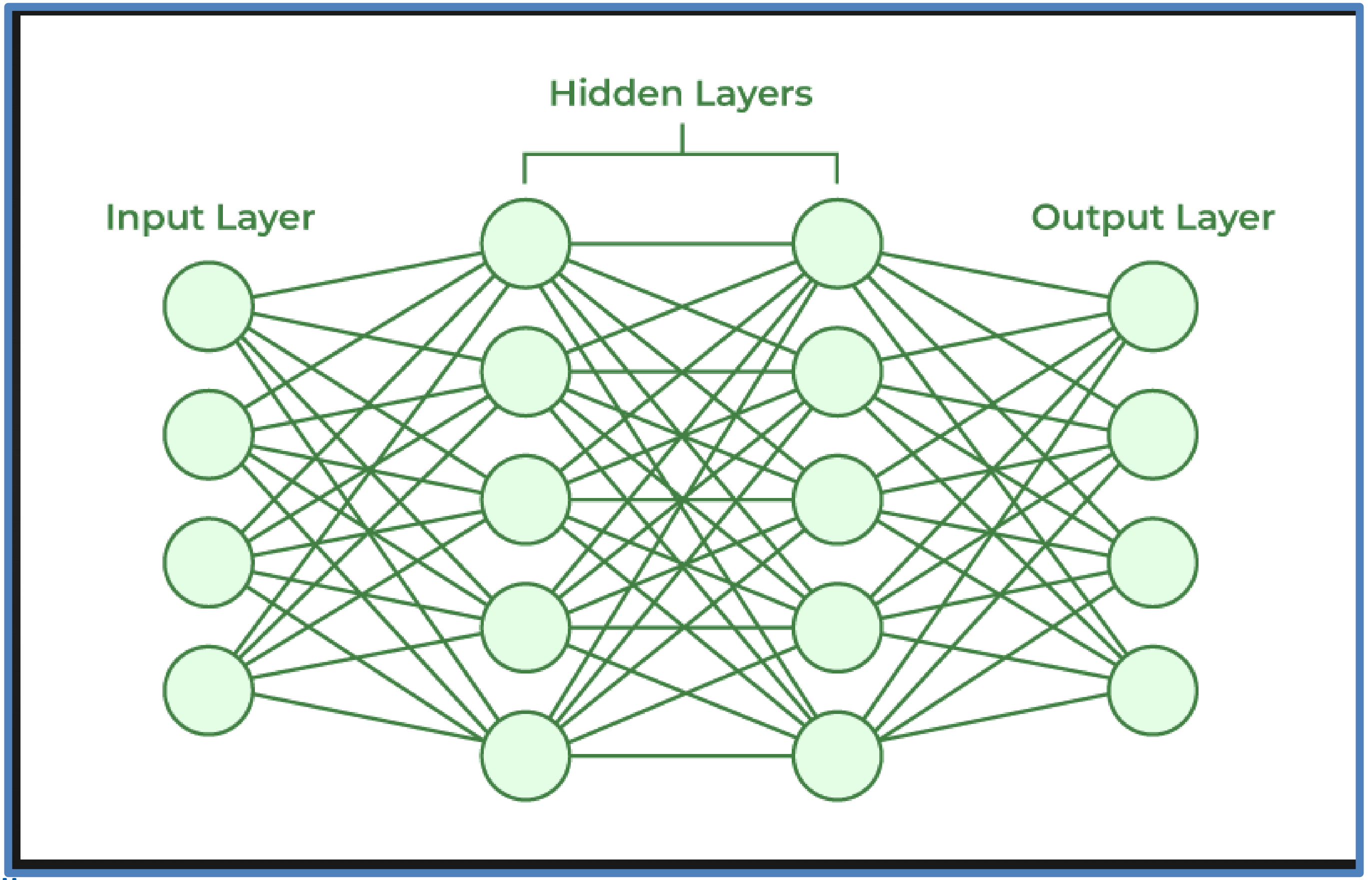
- Therefore, we usually have to assume that it has some restricted form, such as linear

$$Y = X_1 + X_2 + X_3$$

Simple, single layer Path Diagram



Complex Multi-Layer Neural Network Diagram



Linear Regression and Correlation

- This lecture presents methods for analyzing association between quantitative response and explanatory variables.
- The analyses are collectively called a ***regression analysis***.

Example: Regression Analysis for Crime Indices

The table on the next slide shows data from *Statistical Abstract of the United States* for the 50 states and the District of Columbia (D.C.) on

- Murder rate: The number of murders per 100,000 people in the population.
- Violent crime rate: The number of murders, forcible rapes, robberies, and aggravated assaults per 100,000 people in the population.
- Percentage of the population with income below the poverty level.
- Percentage of families headed by a single parent.

Example: Data Set

TABLE 9.1: Statewide Data (from `Crime2` Data File at the Text Website) Used to Illustrate Regression Analyses

State	Violent Crime	Murder Rate	Poverty Rate	Single Parent	State	Violent Crime	Murder Rate	Poverty Rate	Single Parent
AK	761	9.0	9.1	14.3	MT	178	3.0	14.9	10.8
AL	780	11.6	17.4	11.5	NC	679	11.3	14.4	11.1
AR	593	10.2	20.0	10.7	ND	82	1.7	11.2	8.4
AZ	715	8.6	15.4	12.1	NE	339	3.9	10.3	9.4
CA	1078	13.1	18.2	12.5	NH	138	2.0	9.9	9.2
CO	567	5.8	9.9	12.1	NJ	627	5.3	10.9	9.6
CT	456	6.3	8.5	10.1	NM	930	8.0	17.4	13.8
DE	686	5.0	10.2	11.4	NV	875	10.4	9.8	12.4
FL	1206	8.9	17.8	10.6	NY	1074	13.38	16.4	12.7
GA	723	11.4	13.5	13.0	OH	504	6.0	13.0	11.4
HI	261	3.8	8.0	9.1	OK	635	8.4	19.9	11.1
IA	326	2.3	10.3	9.0	OR	503	4.6	11.8	11.3
ID	282	2.9	13.1	9.5	PA	418	6.8	13.2	9.6
IL	960	11.42	13.6	11.5	RI	402	3.9	11.2	10.8
IN	489	7.5	12.2	10.8	SC	1023	10.3	18.7	12.3
KS	496	6.4	13.1	9.9	SD	208	3.4	14.2	9.4
KY	463	6.6	20.4	10.6	TN	766	10.2	19.6	11.2
LA	1062	20.3	26.4	14.9	TX	762	11.9	17.4	11.8
MA	805	3.9	10.7	10.9	UT	301	3.1	10.7	10.0
MD	998	12.7	9.7	12.0	VA	372	8.3	9.7	10.3
ME	126	1.6	10.7	10.6	VT	114	3.6	10.0	11.0
MI	792	9.8	15.4	13.0	WA	515	5.2	12.1	11.7
MN	327	3.4	11.6	9.9	WI	264	4.4	12.6	10.4
MO	744	11.3	16.1	10.9	WV	208	6.9	22.2	9.4
MS	434	13.5	24.7	14.7	WY	286	3.4	13.3	10.8
					DC	2922	78.5	26.4	22.1

	Murder~e	Urban	White	HSGrad	Poverty	Single~d
MurderRate	1.0000					
Urban	0.3306*	1.0000				
	0.0190					
White	-0.5991*	-0.2699	1.0000			
	0.0000	0.0580				
HSGrad	-0.5776*	0.0133	0.3561*	1.0000		
	0.0000	0.9270	0.0112			
Poverty	0.6286*	-0.1556	-0.2300	-0.7735*	1.0000	
	0.0000	0.2805	0.1080	0.0000		
inglePare~d	0.7281*	0.1575	-0.4358*	-0.2360	0.4303*	1.0000
	0.0000	0.2747	0.0016	0.0990	0.0018	

Example: Regression Analysis for Crime Indices

- For these quantitative variables, violent crime rate and murder rate are natural response variables.

- We'll treat the poverty rate and percentage of single-parent families as explanatory variables for these responses as we present regression methods in this lecture.

Linear Regression and Correlation

Aspects of regression analysis:

1. We investigate *whether an association exists* between the variables by testing the hypothesis of statistical independence.
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2. We study the *strength of their association* using the *correlation* measure of association.
3. We estimate a *regression equation* that predicts the value of the response variable from the value of the explanatory variable.

Linear Relationships

- We let *y* denote the *response* variable and let *x* denote the *explanatory* variable. We analyze how values of *y* tend to change from one subset of the population to another, as defined by values of *x*.
- For categorical variables, we did this by comparing the conditional distributions of *y* at the various categories of *x*, in a contingency table.
- For quantitative variables, a mathematical formula describes how the conditional distribution of *y* varies according to the value of *x*.

Linear Functions: Interpreting the y-Intercept and Slope

- ***Linear Function*** - The formula $y = \alpha + \beta x$ expresses observations on y as a ***linear function*** of observations on x. The formula has a straight-line graph with ***slope*** β (beta) and ***y-intercept*** α (alpha).

Linear Functions: Interpreting the y-Intercept and Slope

- Each real number x, when substituted into the formula $y = \alpha + \beta x$, yields a distinct value for y.
- In a graph, the horizontal axis, the ***x-axis***, lists the possible values of x. The vertical axis, the ***y-axis***, lists the possible values of y.
- The axes intersect at the point where $x = 0$ and $y = 0$, called the *origin*.

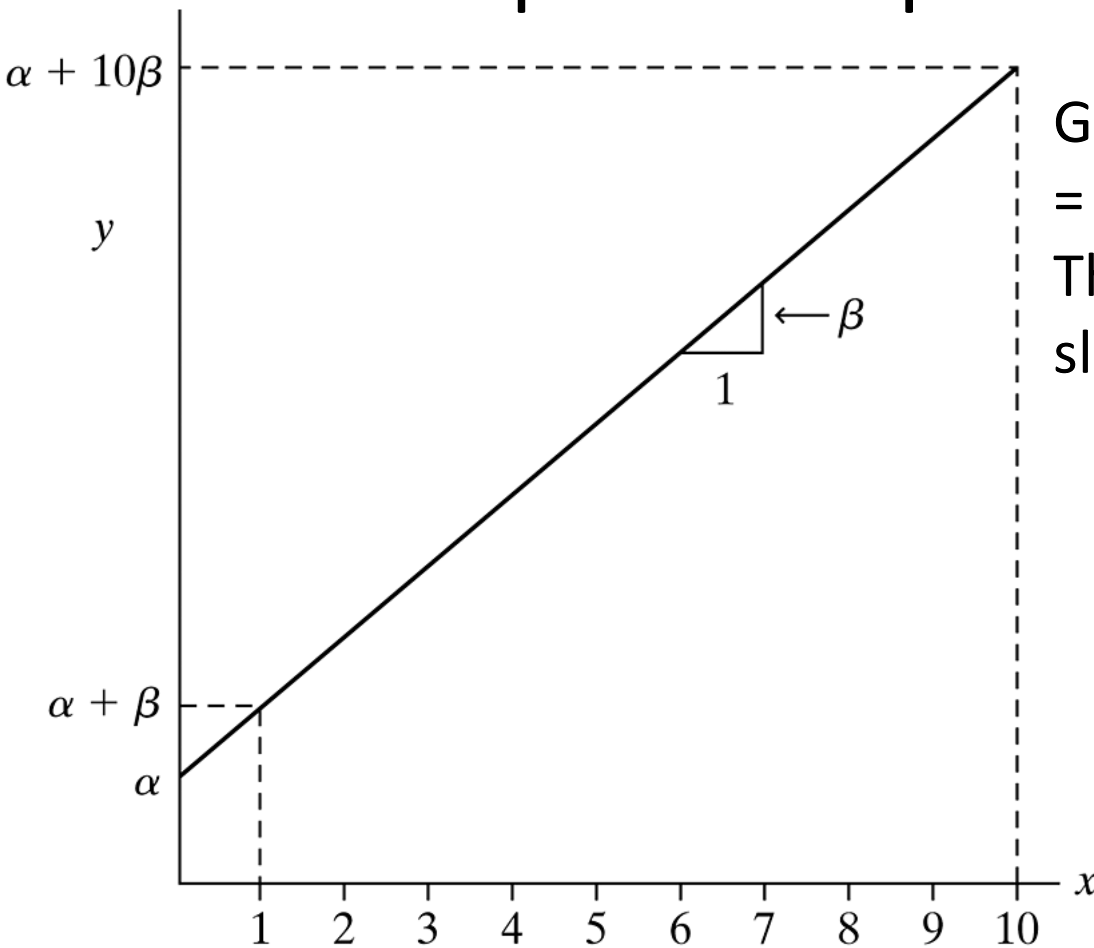
Linear Functions: Interpreting the y-Intercept and Slope

- At $x = 0$, the equation $y = \alpha + \beta x$ simplifies to $y = \alpha + \beta x = \alpha + \beta(0) = \alpha$.
- Thus, the constant α in this equation is the value of y when $x = 0$.
- Points on the y-axis have $x = 0$, so the line has height α at the point of its intersection with the y-axis. Because of this, α is called the ***y-intercept***.

Linear Functions: Interpreting the y-Intercept and Slope

- The ***slope*** *β* equals the change in *y* for a one-unit increase in *x*. For two *x*-values that differ by 1.0 (such as $x = 0$ and $x = 1$), the *y*-values differ by *β*.
- The figure on the next slide portrays the interpretation of the *y*-intercept and slope. In the context of a regression analysis, *α* and *β* are called ***regression coefficients***.

Linear Functions: Interpreting the y-Intercept and Slope



Graph of the Straight Line $y = \alpha + \beta x$.

The y-intercept is α and the slope is β.

Example: Straight Lines for Predicting Violent Crime Rate

For the 50 states, consider y = violent crime rate and x = poverty rate. We'll see that a straight line cannot perfectly represent the relation between them, but the line $y = 210 + 25x$ provides a type of approximation. The y-intercept of 210 represents the violent crime rate at poverty rate $x = 0$. The slope equals 25. When the percentage with income below the poverty level increases by 1, the violent crime rate increases by about 25 crimes a year per 100,000 population.

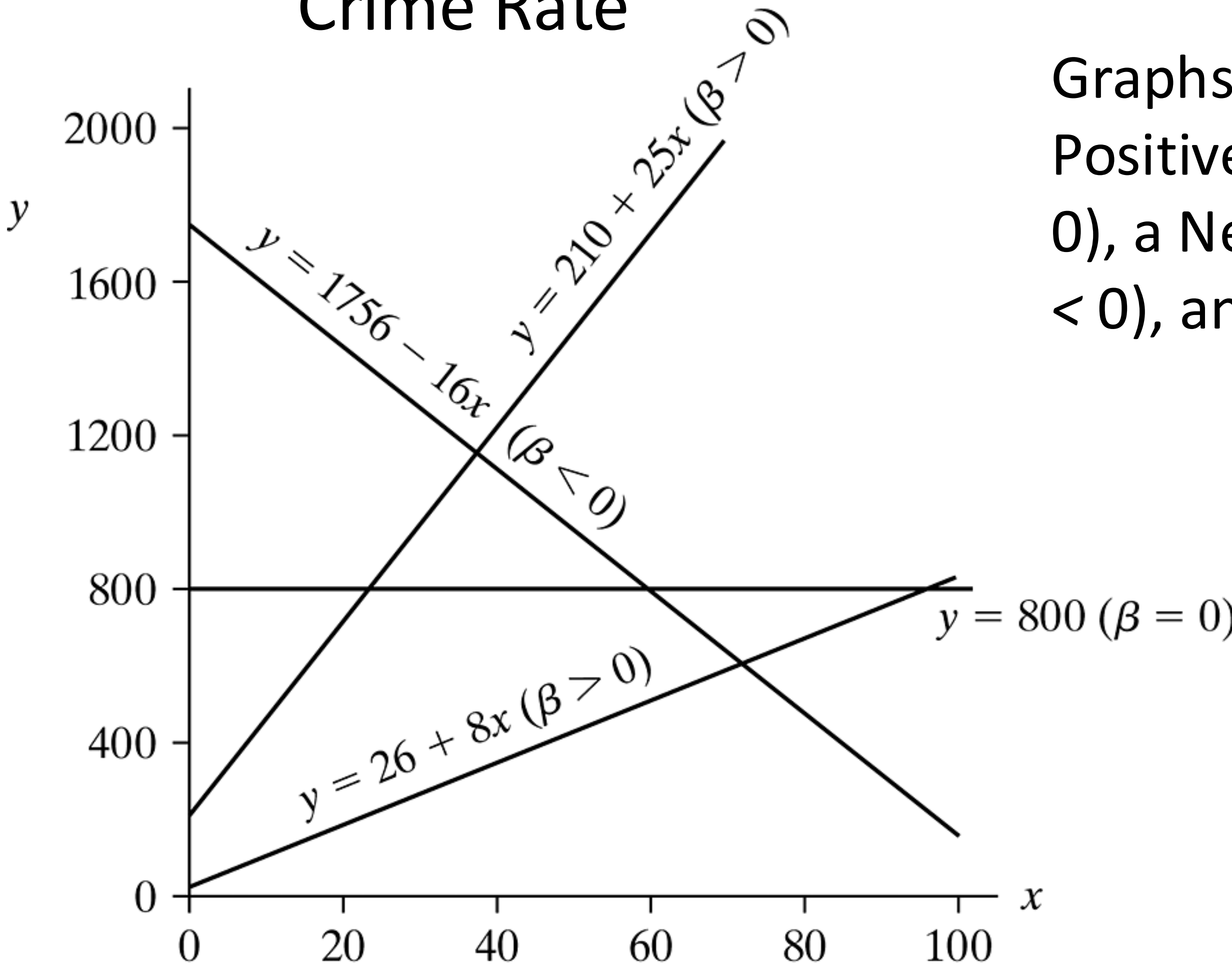
Example: Straight Lines for Predicting Violent Crime Rate

By contrast, if instead x = percentage of the population living in urban areas, the straight line approximating the relationship is $y = 26 + 8x$. The slope of 8 is smaller than the slope of 25 when poverty rate is the explanatory variable. An increase of 1 in the percentage below the poverty level corresponds to a greater change in the violent crime rate than an increase of 1 in the percentage urban.

Example: Straight Lines for Predicting Violent Crime Rate

The figure on the next slide shows the lines relating the violent crime rate to poverty rate and to urban residence. Generally, the larger the absolute value of β, the steeper the line. When β is positive, *y increases* as *x increases*—the straight line goes upward, like these two lines. Then, large values of *y* occur with large values of *x*, and small values of *y* occur with small values of *x*. When a relationship between two variables follows a straight line with $\beta > 0$, the relationship is said to be ***positive***.

Example: Straight Lines for Predicting Violent Crime Rate



Graphs of Lines Showing Positive Relationships (β > 0), a Negative Relationship (β < 0), and Independence (β = 0)

Example: Straight Lines for Predicting Violent Crime Rate

When β is negative, *y decreases* as *x increases*. The straight line then goes downward, and the relationship is said to be ***negative***. For instance, the equation $y = 1756 - 16x$, which has slope −16, approximates the relationship between y = violent crime rate and x = percentage of residents who are high school graduates. For each increase of 1.0 in the percentage who are high school graduates, the violent crime rate decreases by about 16.

Models are Simple Approximations for Reality

- A ***model*** is a simple approximation for the relationship between variables in the population.
- The linear function provides a simple model for the relationship between two quantitative variables.
- For a given value of x, the model $y = \alpha + \beta x$ predicts a value for y.
- The better these predictions tend to be, the better the model.

Models are Simple Approximations for Reality

- *Association does not imply causation*.
- A *sensible* model is actually a bit more complex than the one we've presented so far, by allowing *variability* in *y*-values at each value for *x*. That model, not merely a straight line, is what we mean by a *regression model*.
- We'll next see how to find the best approximating straight line.

9.2 Least Squares Prediction Equation

- Using sample data, we can estimate the equation for the simple straight-line model.
- The process treats α and β in the equation $y = \alpha + \beta x$ as parameters and estimates them.

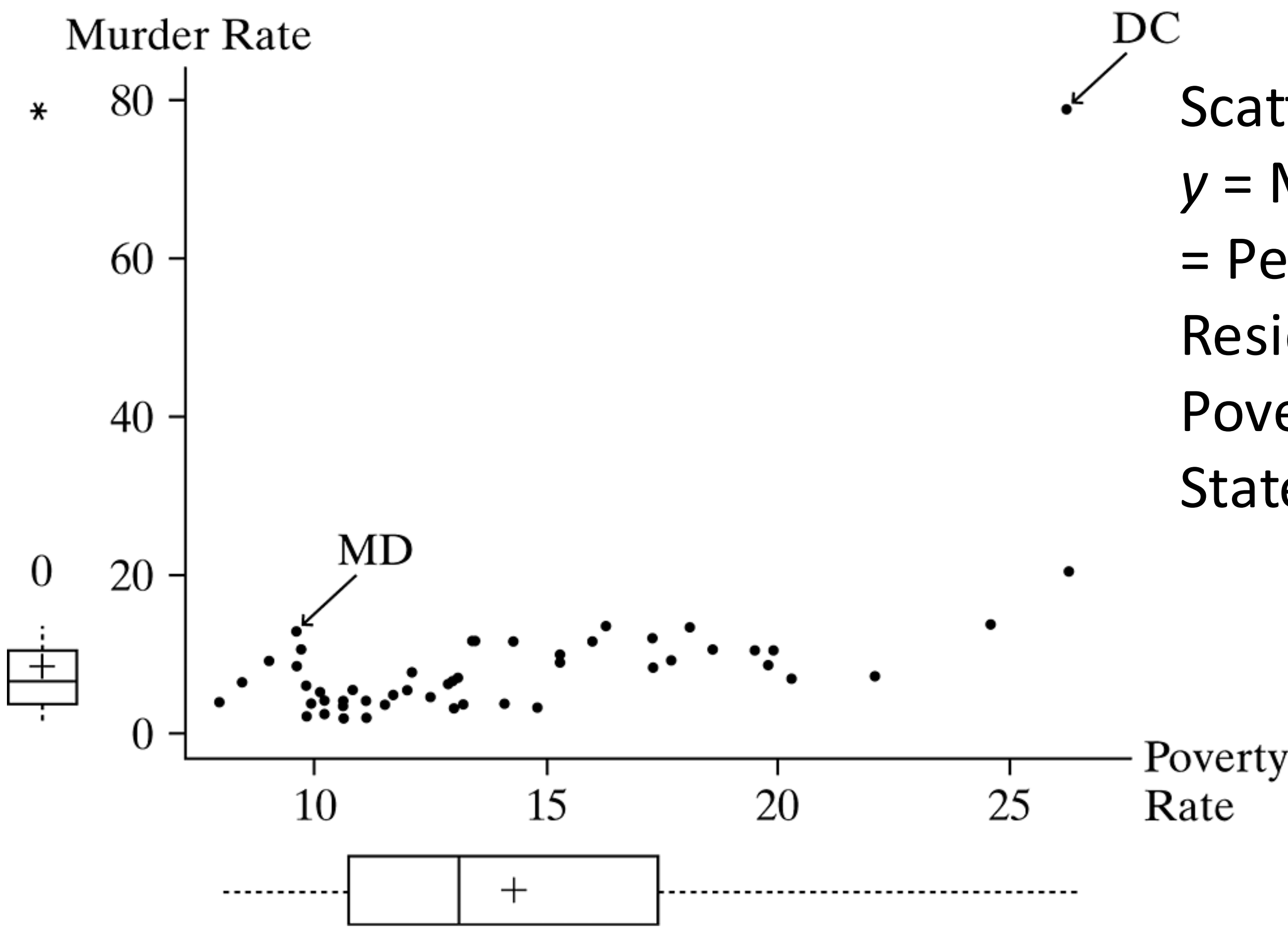
A Scatterplot Portrays the Data

- The first step of model fitting is to plot the data, to reveal whether a model with a straight-line trend makes sense.
- The data values (x, y) for any one subject form a point relative to the x and y axes.
- A plot of the n observations as n points is called a ***scatterplot***.

Example: Scatterplot for Murder Rate and Poverty

For the initial data table provided, let x = poverty rate and
y = murder rate. The figure on the next slide shows a scatterplot for the 51 observations. Each point portrays the values of poverty rate and murder rate for a given state. For Maryland, for instance, the poverty rate is $x = 9.7$, and the murder rate is $y = 12.7$. Its point $(x, y) = (9.7, 12.7)$ has coordinate 9.7 for the x-axis and 12.7 for the y-axis. This point is labeled MD in the figure.

Example: Scatterplot for Murder Rate and Poverty



Scatterplot for *y* = Murder Rate and *x* = Percentage of Residents below the Poverty Level, for 50 States and D.C.

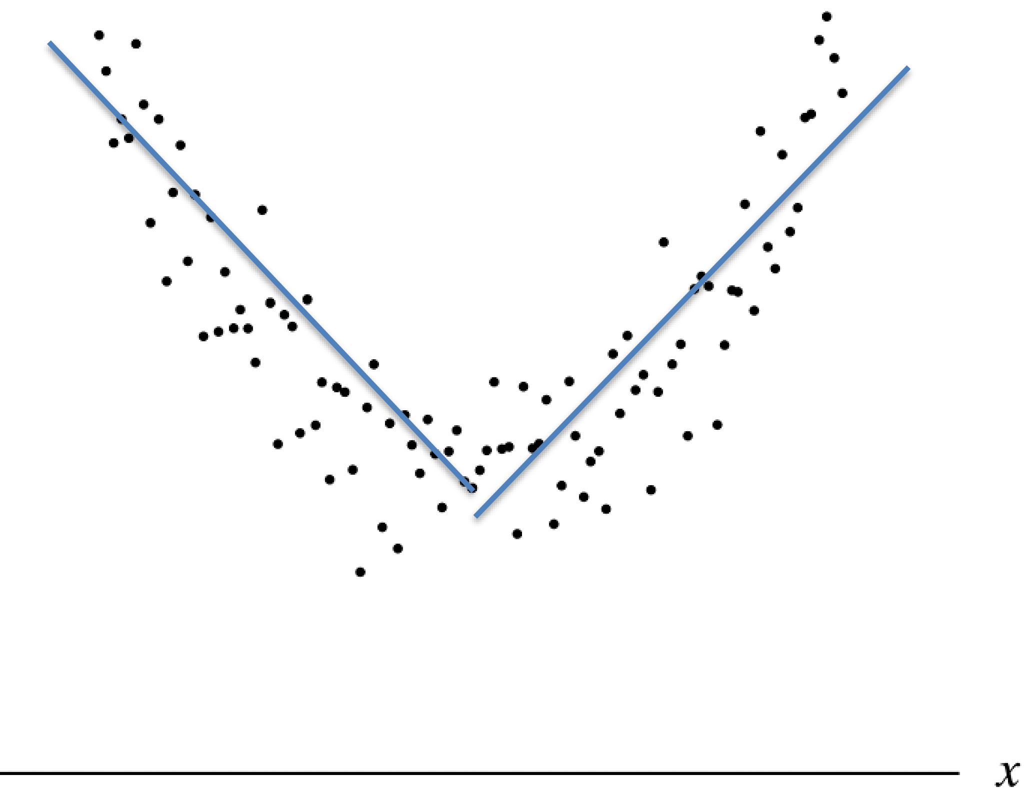
Example: Scatterplot for Murder Rate and Poverty

The figure indicates that the trend of points seems to be approximated fairly well by a straight line. One point, however, is far removed from the rest. This is the point for the District of Columbia (D.C.). It had murder rate much higher than for any state. This point lies far from the overall trend. The figure also shows box plots for these variables. They reveal that D.C. is an extreme *outlier* on murder rate. In fact, it falls 6.5 standard deviations above the mean. We shall see that outliers can have a serious impact on a regression analysis.

A Scatterplot Portrays the Data

- The scatterplot provides a visual check of whether a relationship is approximately linear.
- When the relationship seems highly nonlinear, it is not sensible to use a straight-line model. The next slide’s figure illustrates such a case.

Example: Scatterplot for Murder Rate and Poverty



A Nonlinear Relationship, for Which It Is Inappropriate to Use a Straight-Line Model. Need to fit a curvilinear model.

Prediction Equation

- When the scatterplot suggests that the model $y = \alpha + \beta x$ may be appropriate, we use the data to estimate this line. The notation

$$\hat{y} = a + bx$$

represents a *sample* equation that estimates the linear model. The *y*-intercept (a) estimates the *y*-intercept α of the model and the slope (b) estimates the slope β.

- $\hat{y} = a + bx$ is called the ***prediction equation***, because it provides a prediction $\hat{y}$ for the response variable at any value of x.

Prediction Equation

- The prediction equation is the best straight line, falling closest to the points in the scatterplot, in a sense explained later in this section.
- The formulas for *a* and *b* in the prediction equation $\hat{y} = a + bx$ are

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} \quad \text{and} \quad a = \bar{y} - b\bar{x}.$$

Example: Predicting Murder Rate from Poverty Rate

For the 51 observations on *y* = murder rate and *x* = poverty rate in the first data table, SPSS software provides the results shown in the table below. Murder rate has $\bar{y}$ = 8.7 and *s* = 10.7, indicating that it is

TABLE 9.2: Part of SPSS Output for Fitting Linear Model to Observations from Crime2 Data File for 50 States and D.C. on x = Percentage in Poverty and y = Murder Rate

Variable	Mean	Std Deviation		B	Std. Error
MURDER	8.727	10.718	(Constant)	-10.1364	4.1206
POVERTY	14.259	4.584	POVERTY	1.3230	0.2754

Example: Predicting Murder Rate from Poverty Rate

The estimates of α and β are listed under the heading *B*. The estimated *y*-intercept is $a = -10.14$, listed opposite *(Constant)*. The estimate of the slope is $b = 1.32$, listed opposite the variable name of which it is the coefficient in the prediction equation, *POVERTY*. Therefore, the prediction equation is $\hat{y} = a + bx = -10.14 + 1.32x$.

The slope $b = 1.32$ is positive. So, the larger the poverty rate, the larger is the predicted murder rate. The value 1.32 indicates that an increase of 1 in the percentage living below the poverty rate corresponds to an increase of 1.32 in the predicted murder rate.

Example: Predicting Murder Rate from Poverty Rate

Similarly, an increase of 10 in the poverty rate corresponds to a 10(1.32) = 13.2-unit increase in predicted murder rate. If one state has a 12% poverty rate and another has a 22% poverty rate, for example, the predicted annual number of murders per 100,000 population is 13.2 higher in the second state than the first state. This differential of 13 murders per 100,000 population translates to 130 per million or 1300 per 10 million population. If the two states each had populations of 10 million, the one with the higher poverty rate would be predicted to have 1300 more murders per year.

Effect of Outliers on the Prediction Equation

- The figure on the next slide plots the prediction equation for murder rate versus poverty rate over the scatterplot. The diagram shows that the observation for D.C. (the sole point in the top-right corner) is a ***regression outlier***—it falls quite far from the trend that the rest of the data follow.
- This observation seems to have a substantial effect. The line seems to be pulled up toward it and away from the center of the general trend of points.

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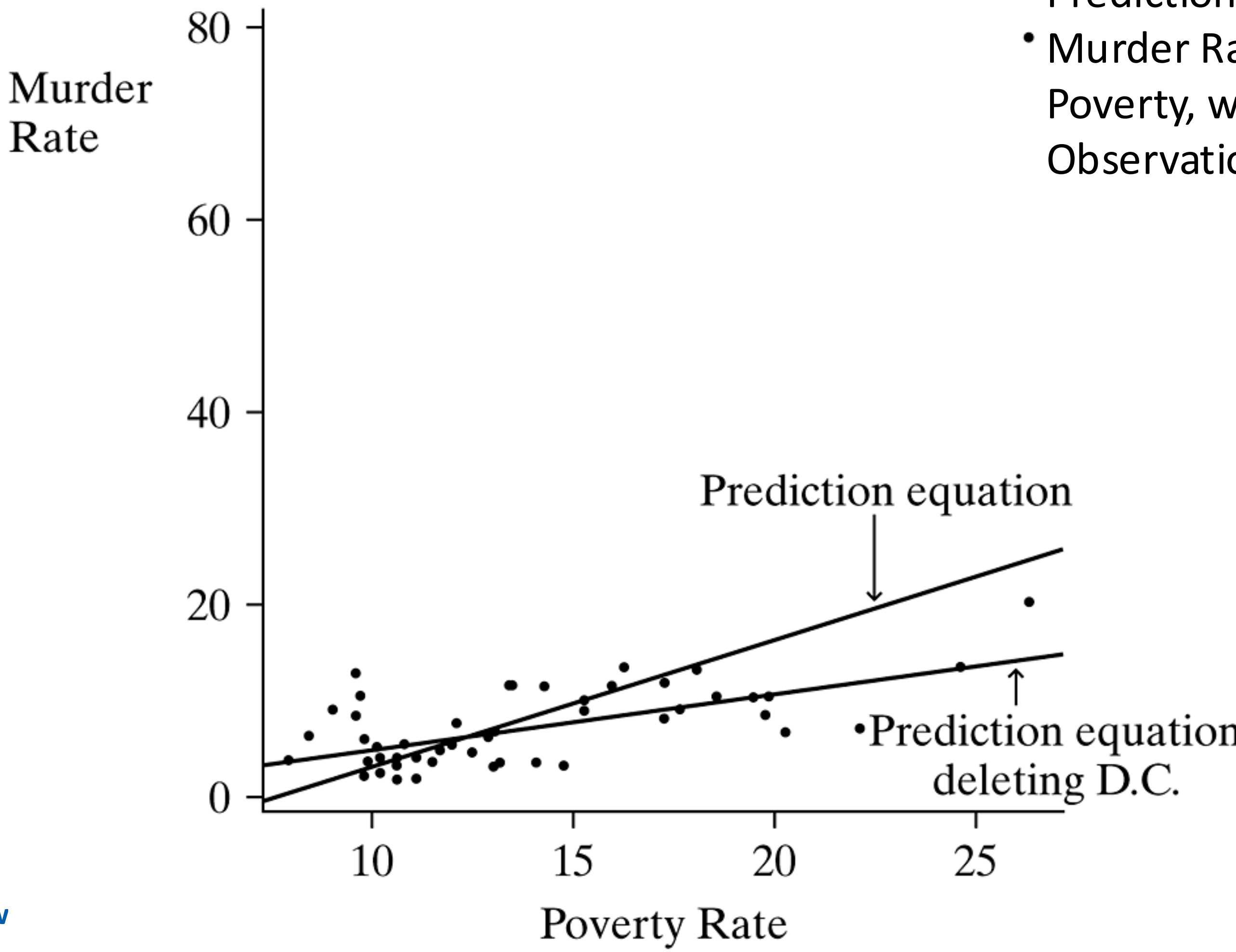
Effect of Outliers on the Prediction Equation



- Prediction Equations Relating Murder Rate and Percentage in Poverty, with and without D.C. Observation

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Effect of Outliers on the Prediction Equation

- Let's now refit the line using the observations for the 50 states but not the one for D.C. The table on the next slide shows that the prediction equation is $\hat{y} = -0.86 + 0.58x$. The previous figure also shows this line, which passes more directly through the 50 points.
- The slope is 0.58, compared to 1.32 when the observation for D.C. is included. The one outlying observation has the impact of more than doubling the slope!

Effect of Outliers on the Prediction Equation

TABLE 9.3: Software Output for Fitting Linear Model to Crime2 Data File on 50 States (but Not D.C.) on x = Percentage in Poverty and y = Murder Rate

	Sum of Squares	df	Mean Square
Regression	307.342	1	307.34
Residual	470.406	48	9.80
Total	777.749	49	

Unstandardized Coefficients	B
(Constant)	-.857
poverty	.584

	murder	predict	residual
1	9.0000	4.4599	4.5401
2	11.6000	9.3091	2.2909
3	10.2000	10.8281	-0.6281
4	8.6000	8.1406	0.4594

Effect of Outliers on the Prediction Equation

- An observation is called ***influential*** if removing it results in a large change in the prediction equation.
- Unless the sample size is large, an observation can have a strong influence on the slope if its *x*-value is low or high compared to the rest of the data and if it is a regression outlier.

Effect of Outliers on the Prediction Equation

- The line for the data set including D.C. seems to distort the relationship for the 50 states. It seems wiser to use the equation based on the 50 states alone rather than to use a single equation for both the 50 states and D.C.
- This line for the 50 states better represents the overall trend. In reporting these results, we would note that the murder rate for D.C. falls outside this trend, being much larger than this equation predicts.

Prediction Errors are Called Residuals

- For the prediction equation $\hat{y} = -0.86 + 0.58x$ for the 50 states, a comparison of the *actual* murder rates to the *predicted* values checks the goodness of the equation.
- Massachusetts had poverty rate $x = 10.7$ and $y = 3.9$. The predicted murder rate ($\hat{y}$) at $x = 10.7$ is $\hat{y} = -0.86 + 0.58x = -0.86 + 0.58(10.7) = 5.4$.
- The prediction error is the difference between the actual y-value of 3.9 and the predicted value of 5.4, or $y - \hat{y} = 3.9 - 5.4 = -1.5$.

Prediction Errors are Called Residuals

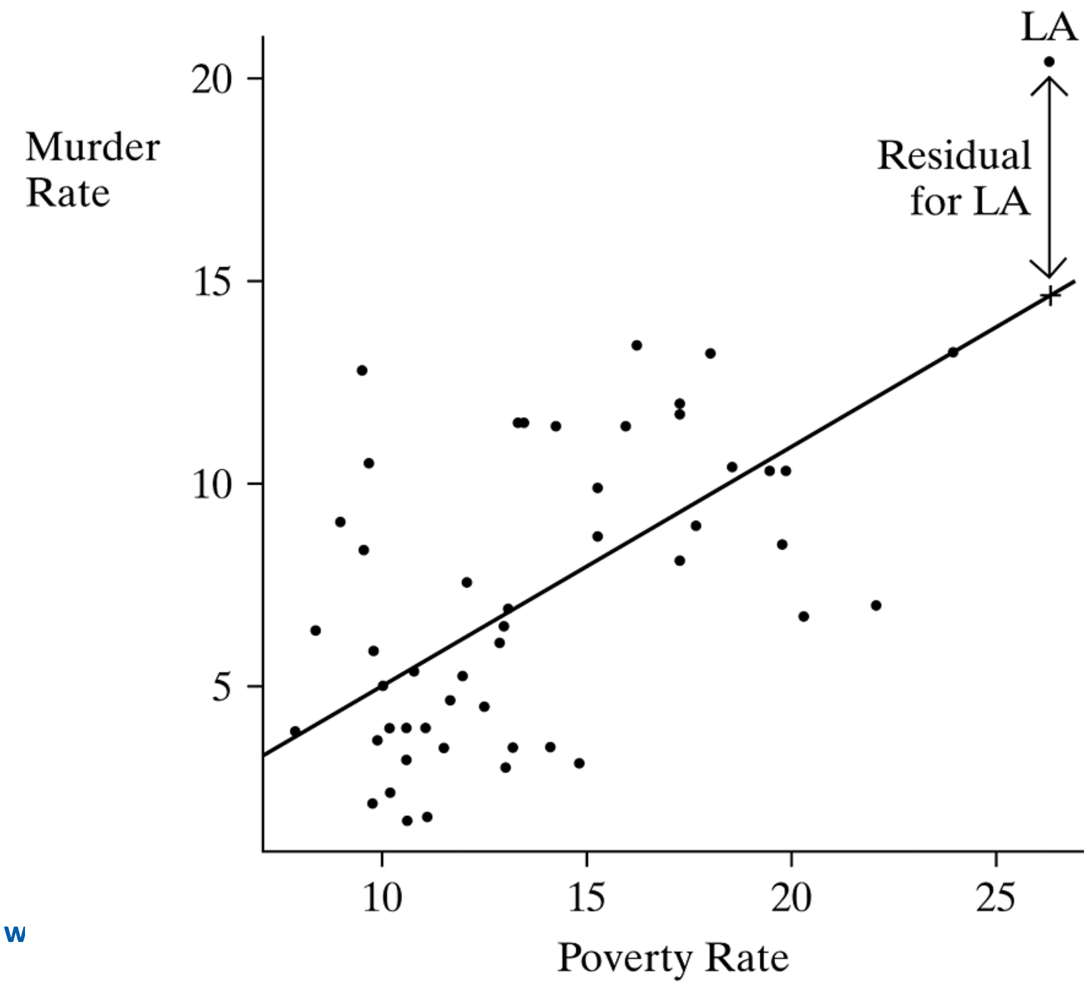
- The prediction errors are called ***residuals***.
- ***Residuals*** - For an observation, the difference between an observed value and the predicted value of the response variable, $y - \hat{y}$, is called the ***residual***.
- The smaller the absolute value of the residual, the better is the prediction, since the predicted value is closer to the observed value.

Prediction Equation has Least Squares Property

- We summarize the size of the residuals by the sum of their squared values. This quantity, denoted by SSE, is

$$\text{SSE} = \sum (y - \hat{y})^2.$$

Prediction Equation has Least Squares Property



Prediction Equation and Residuals for LA. A residual is a vertical distance between a data point and the prediction line.

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Prediction Equation has Least Squares Property

- The residual is computed for every observation in the sample, each residual is squared, and then SSE is the sum of these squares.
- The symbol SSE is an abbreviation for ***sum of squared errors***. This terminology refers to the residual being a measure of prediction error from using $\hat{y}$ to predict y.

Prediction Equation has Least Squares Property

- Least Squares Estimates - The ***least squares estimates*** *a* and *b* are the values that provide the prediction equation $\hat{y} = a + bx$ for which the residual sum of squares, SSE = $\sum(y - \hat{y})^2$, is a minimum.
- The prediction line $\hat{y} = a + bx$ is called the ***least squares line***. If we square the residuals for the least squares line $\hat{y} = -0.86 + 0.58x$ and then sum them, we get

Prediction Equation has Least Squares Property

- Most software (e.g., R, SPSS, Stata) calls SSE the ***residual sum of squares***.

The least squares line

- Has some positive residuals and some negative residuals, but the sum (and mean) of the residuals equals 0.
- Passes through the point ($\bar{x}$, $\bar{y}$).

Prediction Equation has Least Squares Property

- The first property tells us that the too-low predictions are balanced by the too-high predictions.
- Just as deviations of observations from their mean $\bar{y}$ satisfy $\sum(y-\bar{y})=0$, so does the prediction equation satisfy $\sum(y-\hat{y})=0$.
- The second property tells us that the line passes through the center of the data.

The Linear Regression Model

- For the linear model $y = \alpha + \beta x$, each value of x corresponds to a single value of y. Such a model is said to be ***deterministic***.
- It is unrealistic in social science research, because we do not expect all subjects who have the same x-value to have the same y-value. Instead, the y-values *vary*.

The Linear Regression Model

- For example, let x = number of years of education and y = annual income.
- The subjects having $x = 12$ years of education do not all have the same income, because income is not completely dependent upon education. A probability distribution describes annual income for individuals with $x = 12$.
- It is the ***conditional distribution*** of the y-values at $x = 12$.
- A separate conditional distribution applies for those with $x = 13$ years of education.

The Linear Regression Model

- A ***probabilistic*** model for the relationship allows for variability in *y* at each value of *x*.
- We now show how a linear function is the basis for a probabilistic model.

Linear Regression Function

- A probabilistic model uses $\alpha + \beta x$ to represent the *mean* of *y*-values, rather than *y* itself, as a function of *x*. For a given value of *x*, $\alpha + \beta x$ represents the mean of the conditional distribution of *y* for subjects having that value
  of *x*.
- ***Expected Value of y*** - Let $E(y)$ denote the mean of a conditional distribution of *y*. The symbol *E* represents *expected value*.

Linear Regression Function

- We now use the equation $E(y) = \alpha + \beta x$ to model the relationship between x and the mean of the conditional distribution of y.
- An equation of the form $E(y) = \alpha + \beta x$ that relates values of x to the mean of the conditional distribution of y is called a *regression function*.
- ***Regression Function*** - A ***regression function*** is a mathematical function that describes how the mean of the response variable changes according to the value of an explanatory variable.

Linear Regression Function

- The function $E(y) = \alpha + \beta x$ is called a *linear regression function*, because it uses a straight line to relate the mean of y to the values of x.
- Least squares provides the sample prediction equation $\hat{y} = a + bx$.
- At any particular value of x, $\hat{y} = a + bx$ *estimates* the mean of y for all subjects in the population having that value of x.

Describing Variation about the Regression Line

- The linear regression model has an additional parameter σ describing the standard deviation of each conditional distribution.
- σ measures the variability of the *y*-values for all subjects having the same *x*-value.
- We refer to σ as the ***conditional standard deviation***.

Describing Variation about the Regression Line

- A model also assumes a particular probability distribution for the conditional distribution of y. This is needed to make inference about the parameters.
- For quantitative variables, the most common assumption is that the conditional distribution of y is normal at each fixed value of x, with unknown standard deviation σ.

Example: Describing How Income Varies, for Given Education

Suppose $E(y) = -5000 + 3000x$ describes the relationship between mean annual income and number of years of education. The slope $\beta = 3000$ implies that mean income increases \$3000 for each year increase in education. Suppose also that the conditional distribution of income is normal, with $\sigma = 13{,}000$. According to this model, for individuals with x years of education, their incomes have a normal distribution with a mean of $E(y) = -5000 + 3000x$ and a standard deviation of 13,000.

Example: Describing How Income Varies, for Given Education

Those having a high school education (x = 12) have a mean income of $E(y) = -5000 + 3000(12) = 31{,}000$ dollars and a standard deviation of 13,000 dollars. So, about 95% of the incomes fall within two standard deviations of the mean, that is, between 31,000 − 2(13,000) = 5000 and 31,000 + 2(13,000) = 57,000 dollars. Those with a college education (x = 16) have a mean annual income of $E(y) = -5000+3000(16) =$ 43,000 dollars, with about 95% of the incomes falling between \$17,000 and \$69,000. The next figure pictures this regression model.

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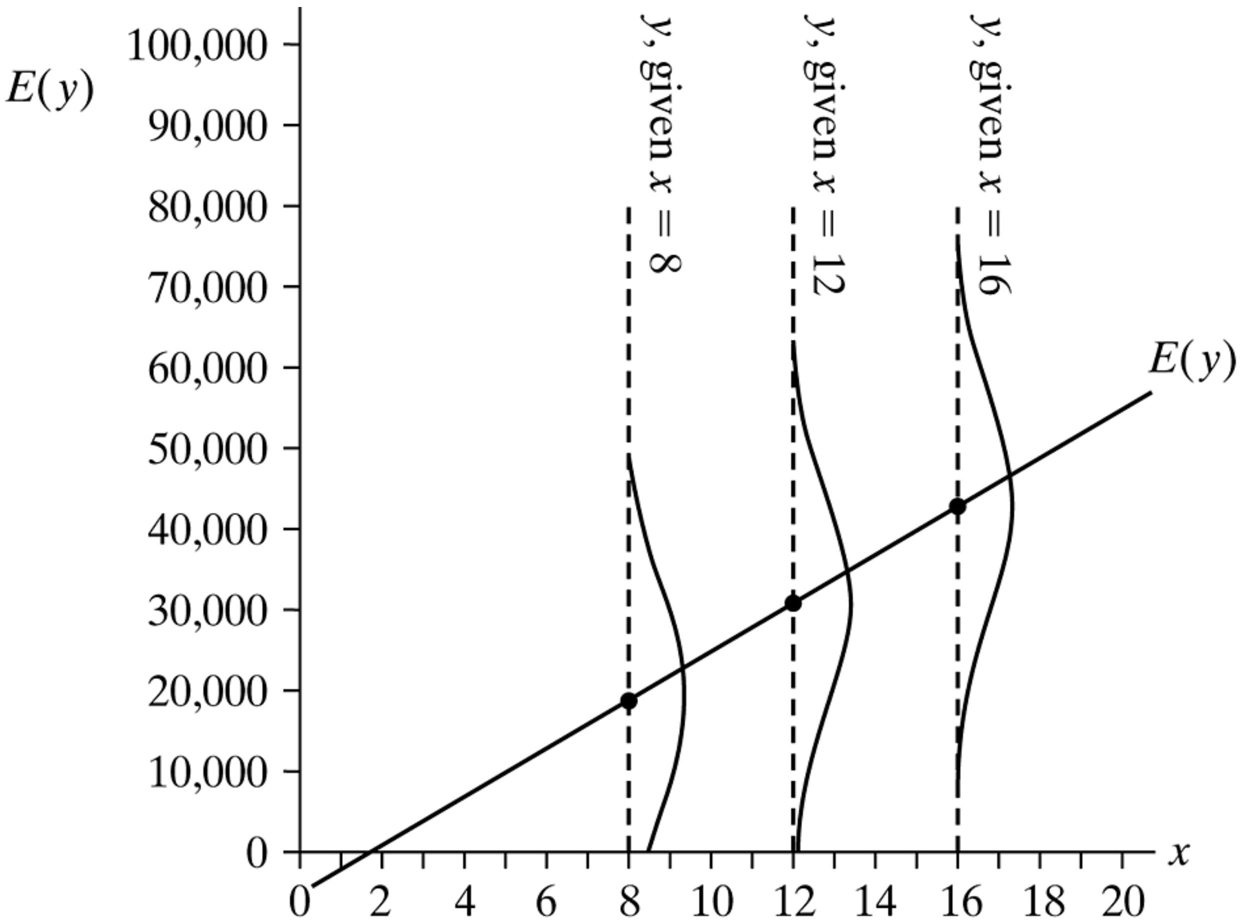
Example: Describing How Income Varies, for Given Education



The Regression Model $E(y) = -5000 + 3000x$, with $\sigma = 13$, Relating the Mean of y = Income (in Dollars) to x = Education (in Years) with conditional income distributions at $x = 8$, 12, and 16 years.

Describing Variation about the Regression Line

- Each conditional distribution is normal, and each has the same standard deviation, $\sigma = 13$.
- In practice, the distributions would not be exactly normal, and the standard deviation need not be the same for each.
- *Any model never holds exactly in practice*. It is merely a simple approximation for reality.

Residual Mean Square: Estimation Conditional Variation

- The estimate of σ uses SSE = $(y - \hat{y})^2$, which measures sample variability about the least squares line. The estimate is

$$s = \sqrt{\frac{\text{SSE}}{n-2}} = \sqrt{\frac{\sum (y - \hat{y})^2}{n-2}}.$$

- If the constant variation assumption is not valid, then s summarizes the *average* variability about the line.

Example: TV Watching and Grade Point Averages

A survey of 50 college students in an introductory psychology class observed self-reports of y = high school GPA and x = weekly number of hours viewing television. The study reported $\hat{y} = 3.44 - 0.03x$. Software reports sums of squares shown in the table on the next slide. This type of table is called an ***ANOVA table***. Here, ANOVA is an acronym for *analysis of variability*. The residual sum of squares in using x to predict y was SSE = 11.66. The estimated conditional standard deviation is

$$s = \sqrt{\frac{\text{SSE}}{n-2}} = \sqrt{\frac{11.66}{50-2}} = 0.49.$$

Example: TV Watching and Grade Point Averages

TABLE 9.4: Software Output of ANOVA Table for Sums of Squares in Fitting Regression Model to y = High School GPA and x = Weekly TV Watching

	Sum of Squares	df	Mean Square
Regression	3.63	1	3.63
Residual	11.66	48	.24
Total	15.29	49	

At any fixed value x of TV viewing, the model predicts that GPAs vary around a mean of $3.44 - 0.03x$ with a standard deviation of 0.49. At $x = 20$ hours a week, for instance, the conditional distribution of GPA is estimated to have a mean of $3.44 - 0.03(20) = 2.84$ and standard deviation of 0.49.

Describing Variation about the Regression Line

- The term ($n - 2$) in the denominator of s is the ***degrees of freedom*** (df) for the estimate.
- When a regression equation has p unknown parameters, then $df = n - p$.
- The table in the above example lists SSE = 11.66 and its $df = n - 2 = 50 - 2 = 48$. The ratio of these, $s^2 = 0.24$, is listed on the printout in the *Mean Square* column.
- It is the *Residual Mean Square*.

Conditional Variation Tends to be Less Than Marginal Variation

- A point estimate of the population standard deviation of a variable *y* is

$$\sqrt{\frac{\sum (y - \bar{y})^2}{n - 1}}.$$

- This is the standard deviation of the *marginal* distribution of *y*, because it uses only the *y*-values.
- It differs from the standard deviation of the *conditional* distribution of *y*, for a fixed value of *x*.

Conditional Variation Tends to be Less Than Marginal Variation

- To reflect its conditional form, that standard deviation is sometimes denoted by $s_{y|x}$ for the sample estimate and $\sigma_{y|x}$ for the population.
- For simplicity, we use s and σ.
- The sum of squares $\sum(y - \bar{y})^2$ in the numerator of s_y is called the ***total sum of squares***. In the GPA example, it is 15.29. The marginal standard deviation of GPA is $s_y = \sqrt{15.29/(50 - 1)} = 0.56$.

Conditional Variation Tends to be Less Than Marginal Variation

- Less spread in *y*-values occurs at a fixed value of *x* than totaled over all such values.
- We'll see that the stronger the association between *x* and *y*, the less the conditional variability tends to be relative to the marginal variability.

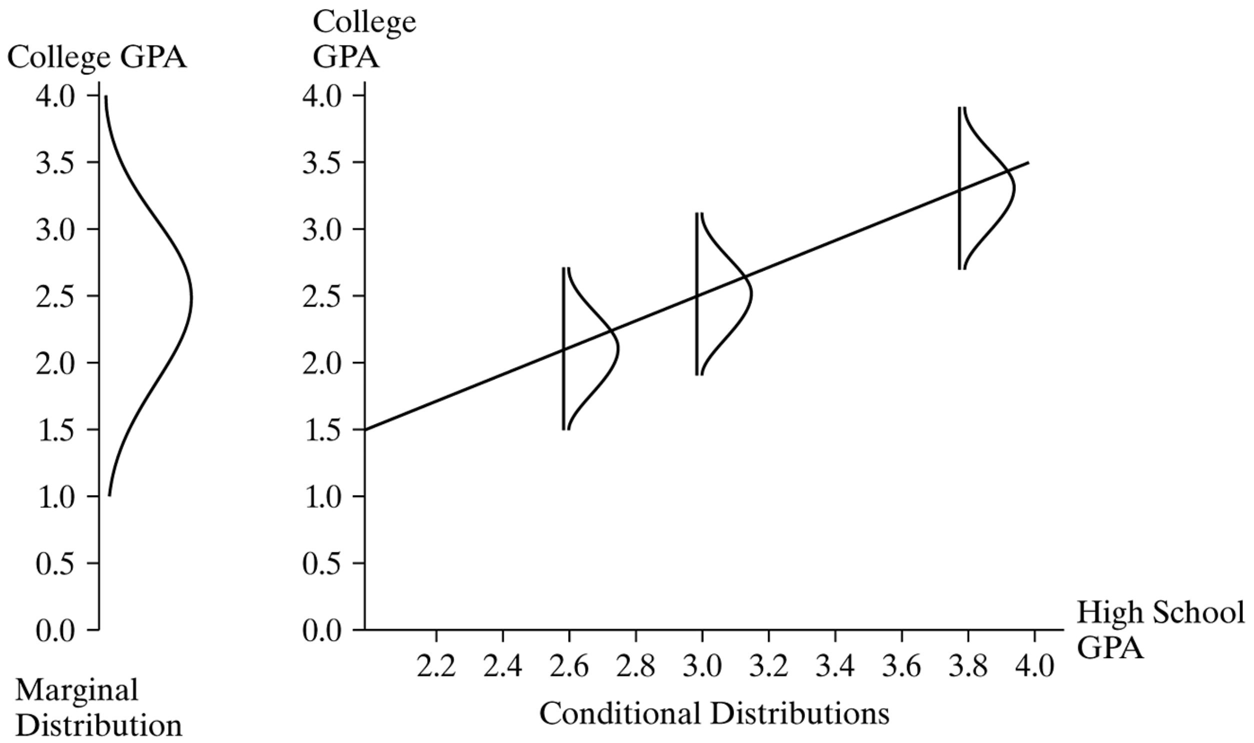
Conditional Variation Tends to be Less Than Marginal Variation

- For example, suppose the *marginal* distribution of college GPAs (y) at your school falls between 1.0 and 4.0, with s_y = 0.60.
- Suppose we could predict college GPA *perfectly* using x = high school GPA, with the prediction equation $\hat{y} = 0.40 + 0.90x$.
- Then, SSE = 0, and the conditional standard deviation would be $s = 0$.

Conditional Variation Tends to be Less Than Marginal Variation

- In practice, perfect prediction would not happen.
- However, the stronger the association in terms of less prediction error, the smaller the conditional variability would be. See the next figure, which portrays a marginal distribution that is much more spread out than each conditional distribution.

Conditional Variation Tends to be Less Than Marginal Variation



Marginal and Conditional Distributions. The marginal distribution shows the overall variability in *y*-values, whereas the conditional distribution shows how *y* varies at a fixed value of *x*.

Measuring Linear Association: The Correlation

- The linear regression model uses a straight line to describe the relationship.
- This section introduces two measures of the strength of association between two quantitative variables.

The Slope and Strength of Association

- The slope b of the prediction equation tells us the *direction* of the association.
- Its sign indicates whether the prediction line slopes upward or downward as x increases.
- The slope does not directly tell us the strength of the association.

The Correlation

- ***Correlation*** - The ***correlation*** between variables *x* and *y*, denoted by *r*, is

$$r = \frac{\sum (x-\bar{x})(y-\bar{y})}{\sqrt{\left[\sum (x-\bar{x})^2\right]\left[\sum (x-\bar{x})^2\right]}}.$$

- The correlation is a *standardized* version of the slope. The standardization adjusts the slope *b* for the fact that the standard deviations of *x* and *y* depend on their units of measurement.

The Correlation

- Let s_x and s_y denote the marginal sample standard deviations of x and y,

$$s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} \text{ and } s_y = \sqrt{\frac{\sum (y - \bar{y})^2}{n-1}}.$$

- ***Correlation is the Standardized Slope*** - The correlation relates to the slope b of the prediction equation $\hat{y} = a + bx$ by

$$r = \left(\frac{s_x}{s_y}\right)b.$$

The Correlation

- Because of the relationship between *r* and *b*, the correlation is also called the ***standardized regression coefficient*** for the model
  $E(y) = \alpha + \beta x$.
- The point estimate *r* of the correlation was proposed by the British statistical scientist Karl Pearson in 1896 and is sometimes called the ***Pearson correlation***.

Example: Correlation Between Murder Rate and Poverty

For the data in first table of the chapter, the prediction equation relating y = murder rate to x = poverty rate is $\hat{y} = -0.86 + 0.58x$. Software tells us that $s_x = 4.29$ for poverty rate, $s_y = 3.98$ for murder rate, and the correlation
$r = 0.63$. In fact,

$$r = \left(\frac{s_x}{s_y}\right)b = \left(\frac{4.29}{3.98}\right)(0.58) = 0.63.$$

We will interpret this value after studying the properties of the correlation.

Properties of the Correlation

- The correlation is valid only when a straight-line model is sensible for the relationship between x and y. Since r is proportional to the slope of a linear prediction equation, it measures the *strength of the linear association*.
- $-1 \leq r \leq 1$. The correlation, unlike the slope b, must fall between −1 and +1. We shall see why later in the section.

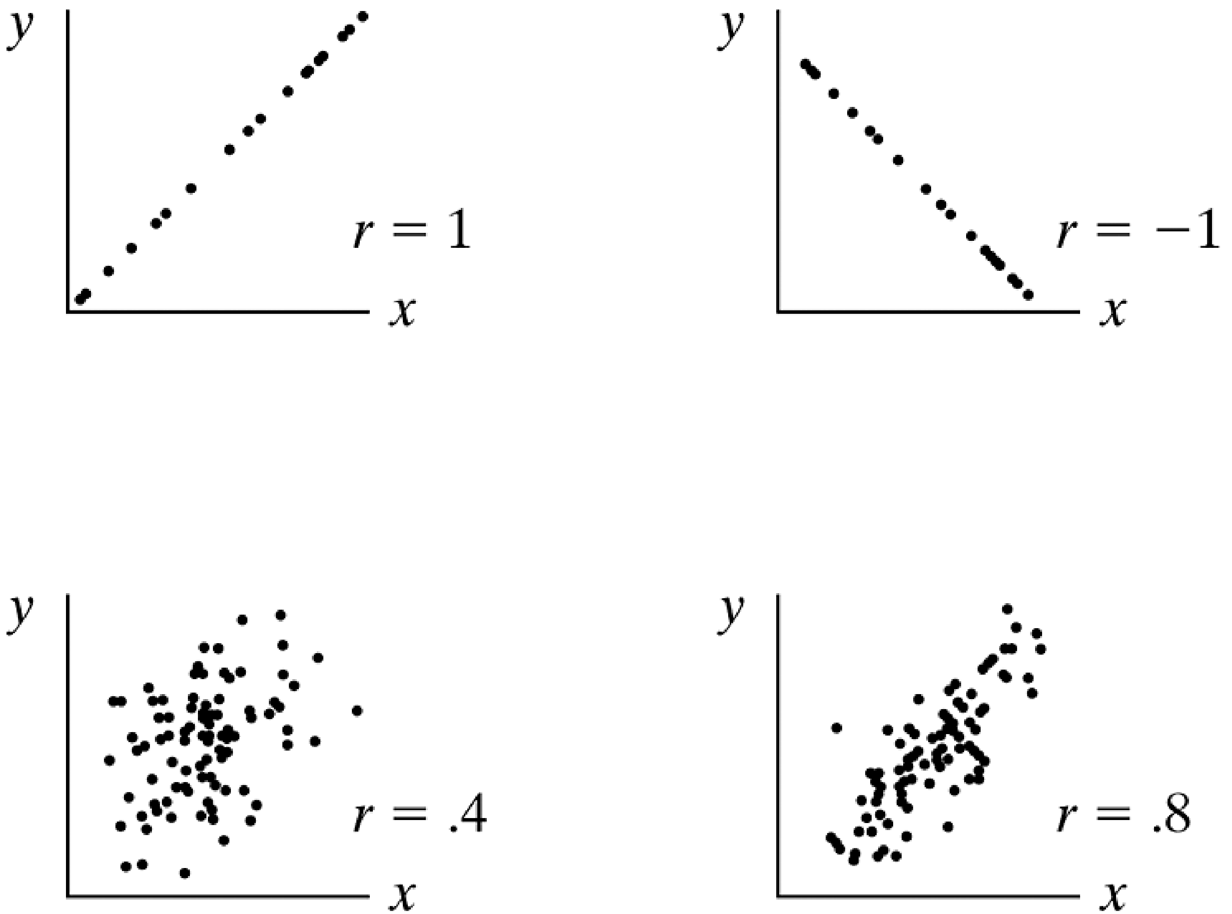
Properties of the Correlation

- *r* has the same sign as the slope *b*. This holds because their formulas have the same numerator, relating to covariation of *x* and *y*, and positive denominators.
- $r = 0$ for those lines having $b = 0$. When $r = 0$, there is not a linear increasing or linear decreasing trend in the relationship.
- $r = \pm 1$ when all the sample points fall exactly on the prediction line. These correspond to *perfect* positive and negative linear associations.

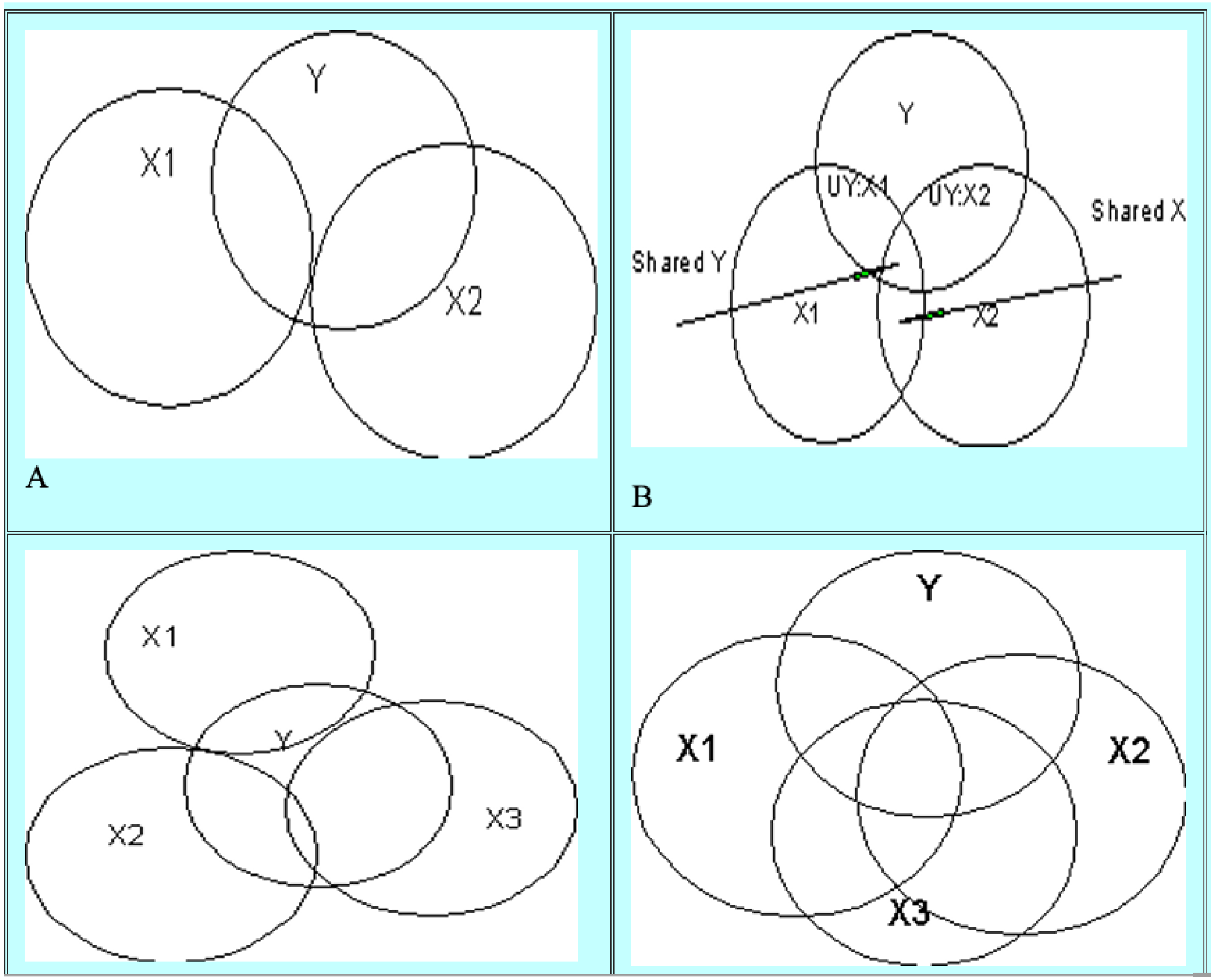
Properties of the Correlation

- The larger the absolute value of *r*, the stronger the linear association. Variables with a correlation of −0.80 are more strongly linearly associated than variables with a correlation of 0.40. The figure on the next slide shows scatterplots having various values for *r*.

Properties of the Correlation



Partial and Semi-Partial Correlations



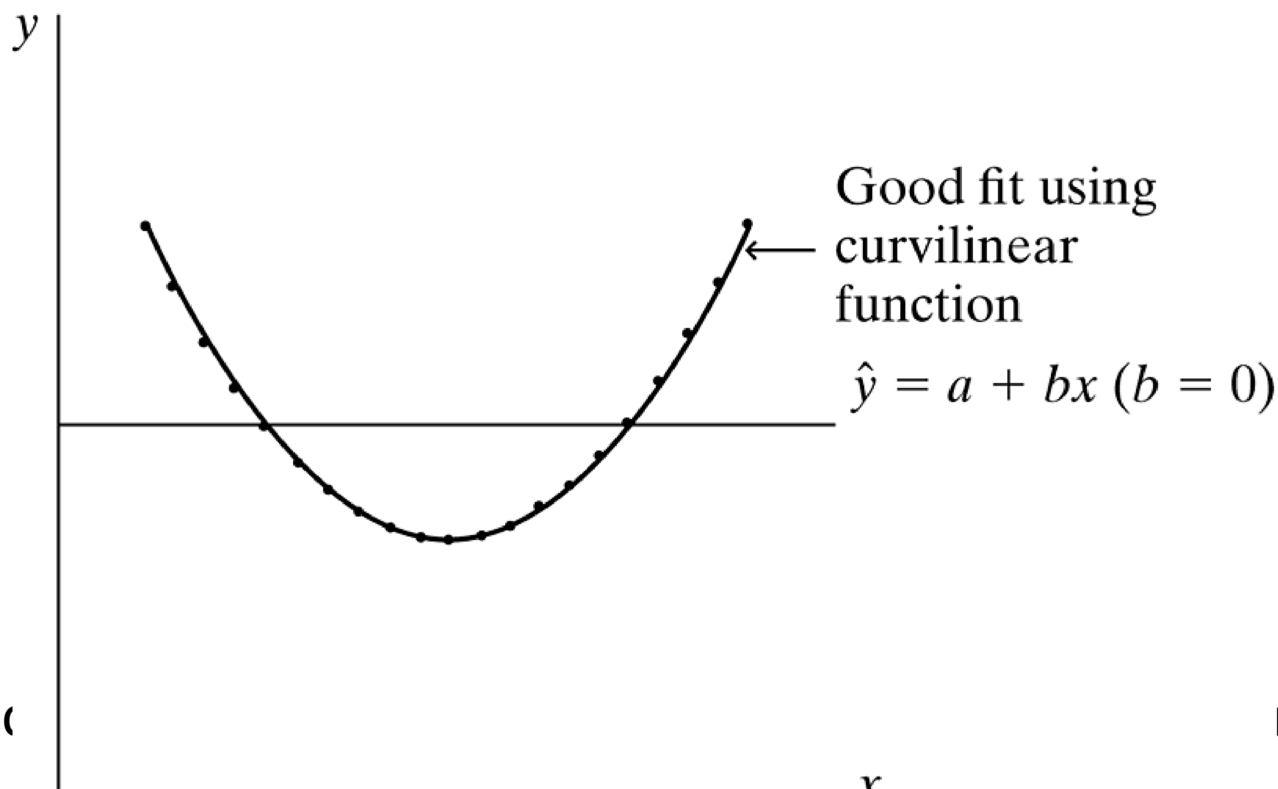
Properties of the Correlation

- The correlation, unlike the slope *b*, treats *x* and *y* symmetrically. The prediction equation using *y* to predict *x* has the same correlation as the one using *x* to predict *y*.
- The value of *r* does not depend on the variables' units.

Properties of the Correlation

- The correlation is useful for comparing associations for variables having different units.
- We emphasize that the correlation describes *linear* relationships. For curvilinear relationships, the best-fitting prediction line may be completely or nearly horizontal, and $r = 0$ when $b = 0$. See the next slide.
- A low absolute value for r does not then imply that the variables are unassociated, but merely that the association is not linear.

Properties of the Correlation



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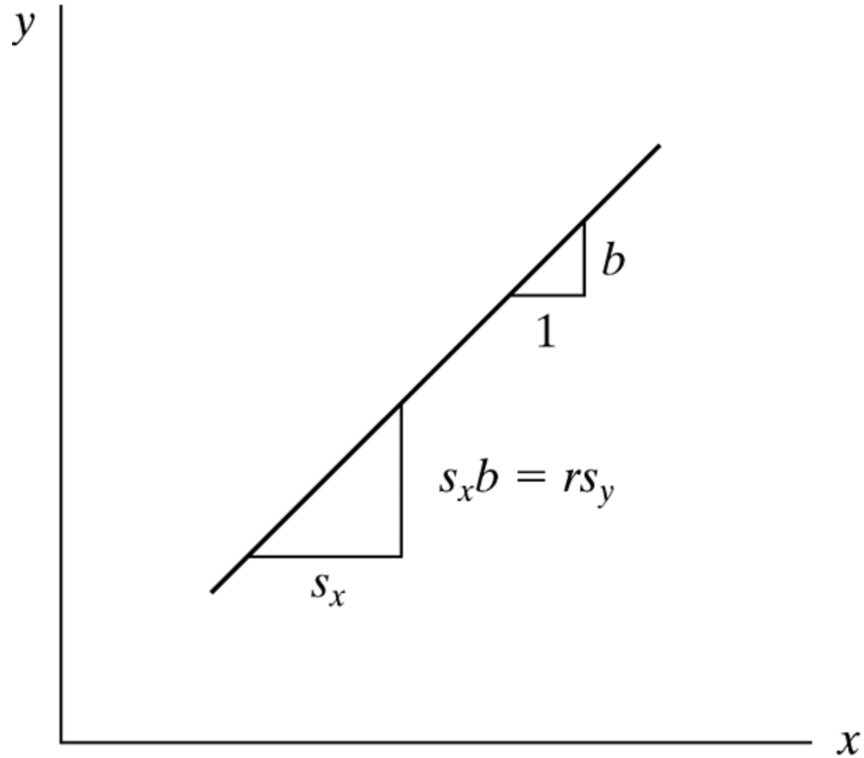
Correlation Implies Regression Toward the Mean

- Another interpretation of the correlation relates to its standardized slope property. We can rewrite the equality

$$r = \left(\frac{s_x}{s_y}\right)b \quad \text{as} \quad s_x b = r s_y.$$

- The slope b is the change in $\hat{y}$ for a one-unit increase in x. An increase in x of s_x units has a predicted change of $s_x b$ units.

Correlation Implies Regression Toward the Mean



An Increase of s_x Units in x Corresponds to a Predicted Change of rs_y Units in y

Correlation Implies Regression Toward the Mean

- Start at the point $(\bar{x}, \bar{y})$ through which the prediction equation passes and consider the impact of x moving above $\bar{x}$ by a standard deviation.
- Suppose that $r = 0.5$. An increase of s_x in x, from $\bar{x}$ to $(\bar{x} + s_x)$, corresponds to a predicted increase of $0.5s_y$ in y, from $\bar{y}$ to $(\bar{y} + 0.5s_y)$. We predict that y is closer to the mean, in standard deviation units.
- This is called ***regression toward the mean***.

Example: Child's Height Regresses Toward the Mean

The British scientist Sir Francis Galton discovered the basic ideas of regression and correlation in the 1880s. After multiplying each female height by 1.08 to account for gender differences, he noted that the correlation between
x = parent height (the average of father's and mother's height) and y = child's height is about 0.5. From the property just discussed, a standard deviation change in parent height corresponds to half a standard deviation change in child's height.

Example: Child's Height Regresses Toward the Mean

For parents of average height, the child's height is predicted to be average. If, on the other hand, parent height is a standard deviation above average, the child is predicted to be half a standard deviation above average. If parent height is two standard deviations below average, the child is predicted to be one standard deviation below average.

Example: Child's Height Regresses Toward the Mean

Since *r* is less than 1, a *y*-value is predicted to be fewer standard deviations from its mean than *x* is from its mean. Tall parents tend to have tall children, but on the average not quite so tall. For instance, if you consider all fathers with height 7 feet, perhaps their sons average 6 feet 5 inches—taller than average, but not so extremely tall; if you consider all fathers with height 5 feet, perhaps their sons average 5 feet 5 inches—shorter than average, but not so extremely short. In each case, Galton pointed out the *regression toward the mean*. This is the origin of the name for regression analysis.

r-Squared: Proportional Reduction in Prediction Error

- A related measure of association summarizes how well x can predict y.
- If we can predict y much better by substituting
  x-values into the prediction equation $\hat{y} = a + bx$ than without knowing the x-values, the variables are judged to be strongly associated.

r-Squared: Proportional Reduction in Prediction Error

This measure of association has four elements:

- A rule for predicting y without using x. We refer to this as Rule 1.
- A rule for predicting y using information on x. We refer to this as Rule 2.
- A summary measure of prediction error for each rule, E_1 for errors by rule 1 and E_2 for errors by rule 2.

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r-Squared: Proportional Reduction in Prediction Error

This measure of association has four elements:

- The difference in the amount of error with the two rules is $E_1 - E_2$. Converting this reduction in error to a proportion provides the definition

$$\text{Proportional reduction in error} = \frac{E_1 - E_2}{E_1}.$$

r-Squared: Proportional Reduction in Prediction Error

- *Rule 1* (Predicting y without using x): The best predictor is $\bar{y}$, the sample mean.
- *Rule 2* (Predicting y using x): When the relationship between x and y is linear, the prediction equation $\hat{y} = a + bx$ provides the best predictor of y.

r-Squared: Proportional Reduction in Prediction Error

- *Prediction Errors:* The prediction error for each subject is the difference between the observed and predicted values of y.
  - The prediction error using rule 1 is $y - \bar{y}$.
  - The prediction error using rule 2 is $y - \hat{y}$, the residual.

r-Squared: Proportional Reduction in Prediction Error

- For each predictor, some prediction errors are positive, some are negative, and the sum of the errors equals 0.
- We summarize the prediction errors by their sum of squared values,

$$E = \sum (\text{observed } y\text{–value} - \text{predicted } y\text{–value}).$$

r-Squared: Proportional Reduction in Prediction Error

- For rule 1, the predicted values all equal $\bar{y}$. The total prediction error is

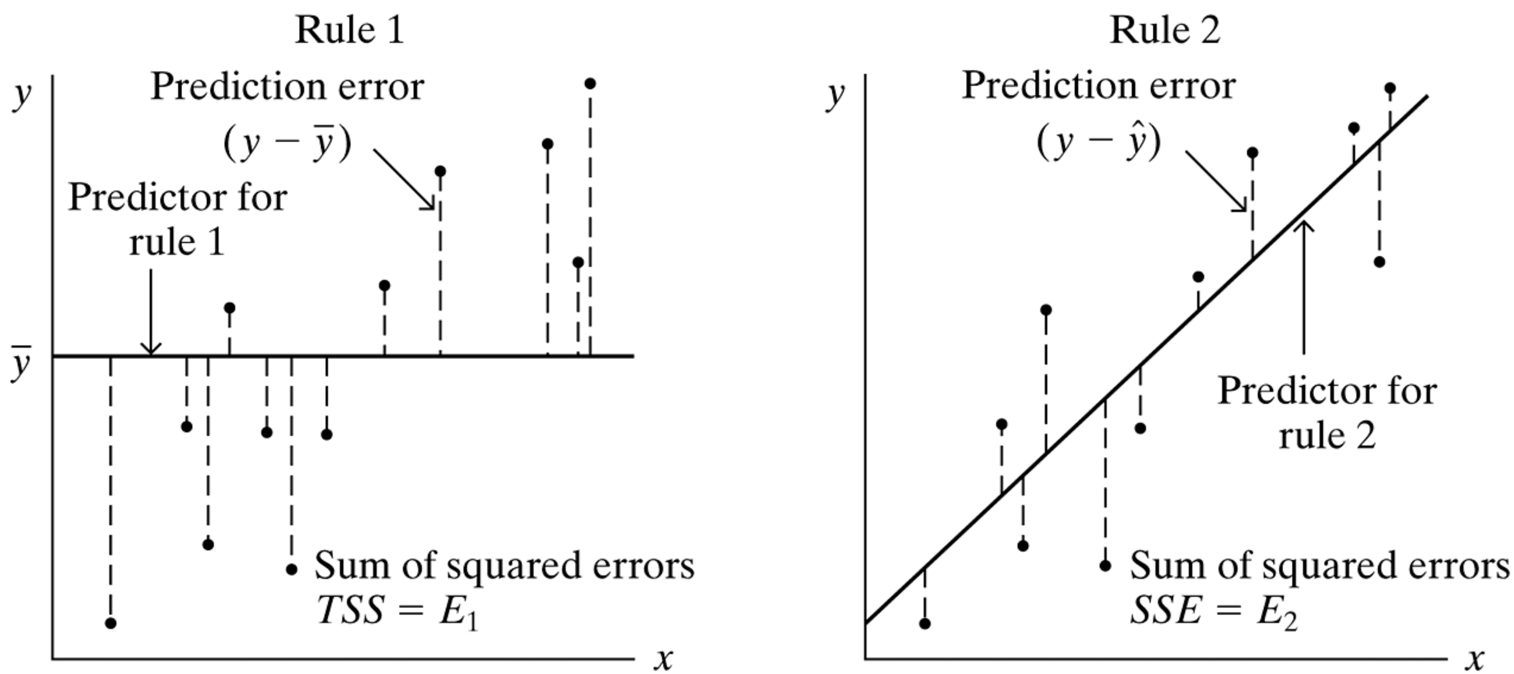
$$E_1 = \sum (y - \bar{y})^2.$$

- This is the *total sum of squares* (TSS) of the y-values about their mean. For rule 2 (predicting using the $\hat{y}$-values), the total prediction error is

$$E_2 = \sum (y - \hat{y})^2.$$

- We have denoted this by SSE, called the ***sum of squared errors*** or the ***residual sum of squares***.

r-Squared: Proportional Reduction in Prediction Error



Graphical Representation of Rule 1 and Total Sum of Squares E_1 = TSS = $\sum (y - \bar{y})^2$, Rule 2 and Residual Sum of Squares E_2 = SSE = $\sum (y - \hat{y})^2$

r-Squared: Proportional Reduction in Prediction Error

- *Definition of Measure:* The proportional reduction in error from using the linear prediction equation instead of $\bar{y}$ to predict y is

$$r^2 = \frac{E_1 - E_2}{E_1} = \frac{\text{TSS} - \text{SSE}}{\text{TSS}} = \frac{\sum (y - \bar{y})^2 - \sum (y - \hat{y})^2}{\sum (y - \bar{y})^2}.$$

- It is called ***r-squared***, or sometimes the ***coefficient of determination***. The notation r^2 is used for this measure because, in fact, the proportional reduction in error equals the square of the correlation r.

Example: r^2 for Murder Rate and Poverty Rate

The correlation between poverty rate and murder rate for the 50 states is $r = 0.629$. Therefore, $r^2 = (0.629)^2 = 0.395$. For predicting murder rate, the linear prediction equation $\hat{y} = -0.86 + 0.58x$ has 39.5% less error than $\bar{y}$.

Software for regression routinely provides tables that contain the sums of squares that compose r^2.
For example, the ANOVA table is below.

	Sum of Squares
Regression	307.342
Residual	470.406
Total	777.749

Example: r^2 for Murder Rate and Poverty Rate

The sum of squared errors using the prediction equation is SSE = $\sum (y - \hat{y})^2 = 470.4$, and the total sum of squares is TSS = $\sum (y - \bar{y})^2 = 777.7$. Thus,

$$r^2 = \frac{\text{TSS} - \text{SSE}}{\text{TSS}} = \frac{777.7 - 470.4}{777.7} = \frac{307.3}{777.7} = 0.395.$$

- In practice, it is unnecessary to perform this computation, since software reports r or r^2 or both.

Property of r-Squared

The properties of r^2 follow directly from those of the correlation r or from its definition in terms of the sums of squares.

- Since $-1 \le r \le 1$, r^2 falls between 0 and 1.
- The minimum possible value for SSE is 0, in which case r^2 = TSS/TSS = 1. For SSE = 0, all sample points must fall exactly on the prediction line. In that case, there is no error using x to predict y with the prediction equation. This condition corresponds to $r = \pm 1$.

Property of r-Squared

- When the least squares slope $b = 0$, the y-intercept a equals $\bar{y}$. Then, $\hat{y} = \bar{y}$ for all x. The two prediction rules are then identical, so
  SSE = TSS and $r^2 = 0$.
- Like the correlation, r^2 measures the strength of *linear* association. The closer r^2 is to 1, the stronger the linear association, in the sense that the more effective the least squares line $\hat{y} = a + bx$ is compared to $\bar{y}$ in predicting y.

Property of *r*-Squared

- r^2 does not depend on the units of measurement, and it takes the same value when x predicts y as when y predicts x.

Sum of Squares Describe Conditional and Marginal Variability

- The total sum of squares, TSS = $\sum(y-\bar{y})^2$, summarizes the *variability* of the observations on *y*, since this quantity divided by *n*−1 is the sample variance s_y^2 of the *y*-values.
- SSE = $\sum(y-\hat{y})^2$ summarizes the variability around the prediction equation, which refers to variability for the conditional distributions.
- When $r^2 = 0.39$, the variability in *y* using *x* to make the predictions is 39% less than the overall variability of the *y*-values.

Inferences for the Slope and Correlation

- A confidence interval for the slope of the regression equation or the correlation tells us about the size of the effect.
- These inferences enable us to judge whether the variables are associated and to estimate the direction and strength of the association.

Assumptions for Statistical Inference

Statistical inferences for regression make the following assumptions:

- Randomization, such as a simple random sample in a survey.
- The mean of *y* is related to *x* by the linear equation $E(y) = \alpha + \beta x$.
- The conditional standard deviation σ is identical at each *x*-value.
- The conditional distribution of *y* at each value of *x* is normal.

Test of Independence Using Slope or Correlation

- Under the assumptions on the previous slide, suppose the normal conditional distribution of *y* is the same at each *x*-value. Then, the two quantitative variables are statistically independent.
- For the linear regression function $E(y) = \alpha + \beta x$, this means that the slope $\beta = 0$. See next slide.
- The null hypothesis that the variables are statistically independent is H_0: $\beta = 0$.

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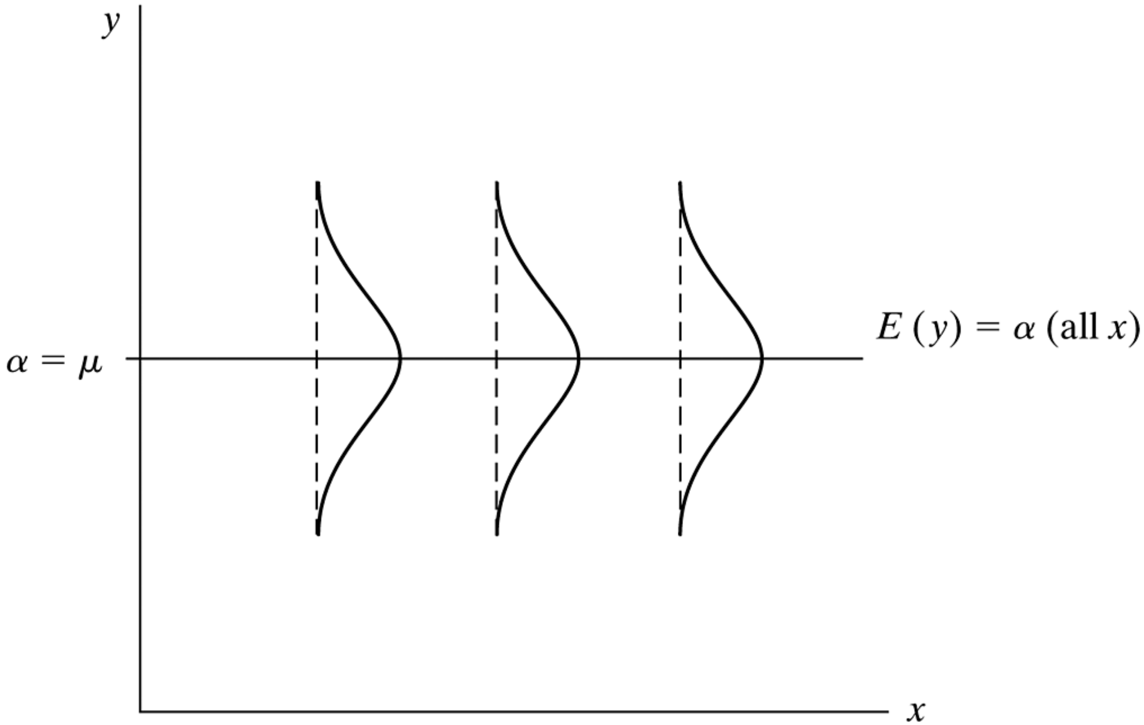
Test of Independence Using Slope or Correlation



x and *y* Are Statistically Independent when the Slope $\beta = 0$ in the Regression Model $E(y) = \alpha + \beta x$ with Normal Conditional Distributions Having Identical Standard Deviations

Test of Independence Using Slope or Correlation

- We can test independence against H_a: $\beta \neq 0$, or a one-sided alternative, H_a:$\beta > 0$ or H_a:$\beta <$ 0, to predict the direction of the association.
- The test statistic is

$$t = \frac{b}{se},$$

where *se* is the standard error of the sample slope *b*.

Test of Independence Using Slope or Correlation

- We take the estimate *b* of the parameter *β*, subtract the null hypothesis value ($\beta = 0$), and divide by the standard error of the estimate *b*.
- This test statistic has the *t* sampling distribution with $df = n - 2$.
- The *P*-value for H_a: $\beta \neq 0$ is the two-tail probability from the *t* distribution.
- The formula for the standard error of *b* is

$$se = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}, \text{ where } s = \sqrt{\frac{\text{SSE}}{n-2}}.$$

Test of Independence Using Slope or Correlation

- The correlation $r = 0$ in the same situations in which the slope $b = 0$.
- Let ρ denote the correlation value in the population. Then, $\rho = 0$ precisely when $\beta = 0$.
- A test of H_0: $\rho = 0$ using the sample value r is equivalent to the t test of H_0: $\beta = 0$ using the sample value b. The test statistic for H_0: $\rho = 0$ is

$$t = \frac{r}{\sqrt{\frac{1 - r^2}{n - 2}}}.$$

Example: Regression for Selling Price of Homes

What affects the selling price of a house? The table on the next slide shows observations on recent home sales in Gainesville, Florida. This table shows data for eight homes. The entire file for 100 home sales is the Houses data file at the text website. Variables listed are selling price (in dollars), size of house (in square feet), annual property taxes (in dollars), number of bedrooms, number of bathrooms, and whether the house is newly built.

Example: Regression for Selling Price of Homes

TABLE 9.5: Selling Prices and Related Factors for a Sample of Home Sales in Gainesville, Florida (Houses Data File)

Home	Selling Price	Size	Taxes	Bedrooms	Bathrooms	New
1	279,900	2048	3104	4	2	No
2	146,500	912	1173	2	1	No
3	237,700	1654	3076	4	2	No
4	200,000	2068	1608	3	2	No
5	159,900	1477	1454	3	3	No
6	499,900	3153	2997	3	2	Yes
7	265,500	1355	4054	3	2	No
8	289,900	2075	3002	3	2	Yes

Note: The complete Houses data file for 100 homes is at the text website.

Example: Regression for Selling Price of Homes

For a set of variables, software can report the correlation for each pair in a ***correlation matrix***. This matrix is a square table listing the variables as the rows and again as the columns. The table shows the way software reports the correlation matrix for the variables selling price, size, taxes, and number of bedrooms.

TABLE 9.6: Correlation Matrix for House Selling Price Data from Houses Data File

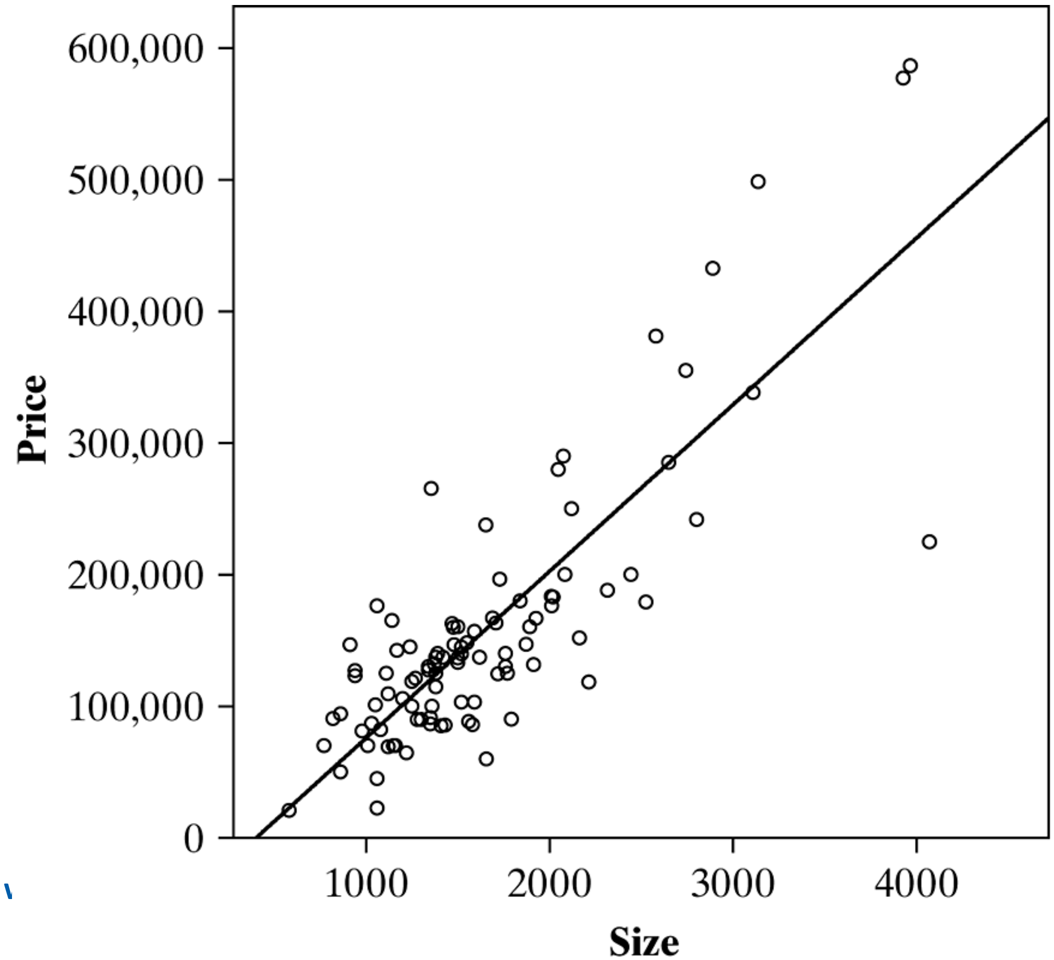
	Correlations			
	price	size	taxes	bedrooms
price	1.00000	0.83378	0.84198	0.39396
size	0.83378	1.00000	0.81880	0.54478
taxes	0.84198	0.81880	1.00000	0.47393
bedrooms	0.39396	0.54478	0.47393	1.00000

Example: Regression for Selling Price of Homes

Use only the data on y = selling price and x = size of house. Since these 100 observations come from one city alone, we cannot use them to make inferences about the relationship between x and y in general. We treat them as a random sample of a conceptual population of home sales in this market in order to analyze how these variables seem to be related.

The figure on the next slide shows a scatterplot, which displays a strong positive trend. The model $E(y) = \alpha + \beta x$ seems appropriate. Some of the points at high levels of size may be outliers, however, and one point falls quite far below the overall trend.

Example: Regression for Selling Price of Homes



Scatterplot and Prediction Equation for *y* = Selling Price (in Dollars) and *x* = Size of House (in Square Feet)

Example: Regression for Selling Price of Homes

The table shows some software output (Stata) for a regression analysis. The prediction equation is $\hat{y} = -50{,}926.2 + 126.6x$. The predicted selling price increases by $b = 126.6$ dollars for an increase in size of a square foot. The previous figure also superimposes the prediction equation over the scatterplot.

TABLE 9.7: Stata Output (Edited) for Regression Analysis of y = Selling Price and x = Size of House from Houses Data File

Source	SS	df	MS
Model	7.0573e+11	1	7.0573e+11
Residual	3.0942e+11	98	3.1574e+09
Total	1.0151e+12	99	1.0254e+10

Number of obs	= 100
F(1, 98)	= 223.52
Prob > F	= 0.0000
R-squared	= 0.6952
Adj R-squared	= 0.6921
Root MSE	= 56190

price	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
size	126.5941	8.467517	14.95	0.000	109.7906	143.3976
_cons	-50926.25	14896.37	-3.42	0.001	-80487.62	-21364.89

Example: Regression for Selling Price of Homes

The previous table reports that the standard error of the slope estimate is se = 8.47. This value estimates the variability in sample slope values that would result from repeatedly selecting random samples of 100 house sales in Gainesville and calculating prediction equations. For testing independence, H_0: $\beta = 0$, the test statistic is

$$t = \frac{b}{se} = \frac{126.6}{8.47} = 14.95.$$

Example: Regression for Selling Price of Homes

Since n = 100, its degrees of freedom are $df = n - 2$ = 98. This is an extremely large test statistic. $P > |t|$, is 0.000 to three decimal places. This refers to the two-sided alternative H_a: $\beta \neq 0$. It is the two-tailed probability of a t statistic at least as large in absolute value as the absolute value of the observed t, $|t| = 14.95$, presuming H_0 is true.

Example: Regression for Selling Price of Homes

We get the same result if we conduct the test using the correlation. The correlation of r = 0.834 for the house selling price data has

$$t = \frac{r}{\sqrt{\frac{1 - r^2}{n - 2}}} = \frac{0.834}{\sqrt{\frac{1 - 0.834^2}{98}}} = 14.95.$$

The evidence is extremely strong that size of house has a positive effect on selling price. On the average, selling price increases as size increases.

Confidence interval for the Slope and Correlation

- A confidence interval for β has the formula

$$b \pm t(se).$$

- The t-score is the value, with $df = n - 2$, for the desired confidence level.
- The form of the interval is similar to the confidence interval for a mean, namely, take the estimate of the parameter and add and subtract a t multiple of the standard error.
- The se is the same as se in the test about β.

Example: Estimating the Slope and Correlation for House Selling Prices

For the data on x = size of house and y = selling price, b = 126.6 and se = 8.47. The parameter β refers to the change in the mean selling price (in dollars) for each 1-square-foot increase in size. For a 95% confidence interval, we use the $t_{.025}$ value for $df = n - 2 = 98$, which is $t_{.025} = 1.984$. The interval is

$$b \pm t_{.025}(se) = 126.6 \pm 1.984(8.47) = 126.6 \pm 16.8,$$

$$\text{or } (110, 143).$$

We can be 95% confident that β falls between 110 and 143. The mean selling price increases by between \$110 and \$143 for a 1-square-foot increase in house size.

Sum of Squares in Software Output

- How do we interpret the sums of squares (SS) output in tables such as the output table for a previous example?
- In that table, the residual sum of squares (SSE = 3.0942×10^{11}) is a huge number because the *y*-values are very large and their deviations are squared.
- The estimated conditional standard deviation of *y* is

$$s = \sqrt{SSE/(n-2)} = 56{,}190.$$

Sum of Squares in Software Output

- The sum of squares table also reports the total sum of squares, TSS = $\sum(y - \bar{y})^2$ = 1.0151×10^{12}. From this value and SSE,

$$r^2 = \frac{\text{TSS} - \text{SSE}}{\text{TSS}} = 0.695.$$

- This is the proportional reduction in error in predicting the selling price using the linear prediction equation instead of the sample mean selling price. A strong association exists between these variables.

Sum of Squares in Software Output

- The total sum of squares TSS partitions into two parts, the sum of squared errors, SSE = 3.0492×10^{11}, and the difference TSS – SEE = 7.0573×10^{11}. This difference is the numerator of the r^2 measure.

- For the difference TSS – SSE, software calls this the ***regression sum of squares*** (e.g., SPSS) or the ***model sum of squares*** (e.g., Stata, SAS).

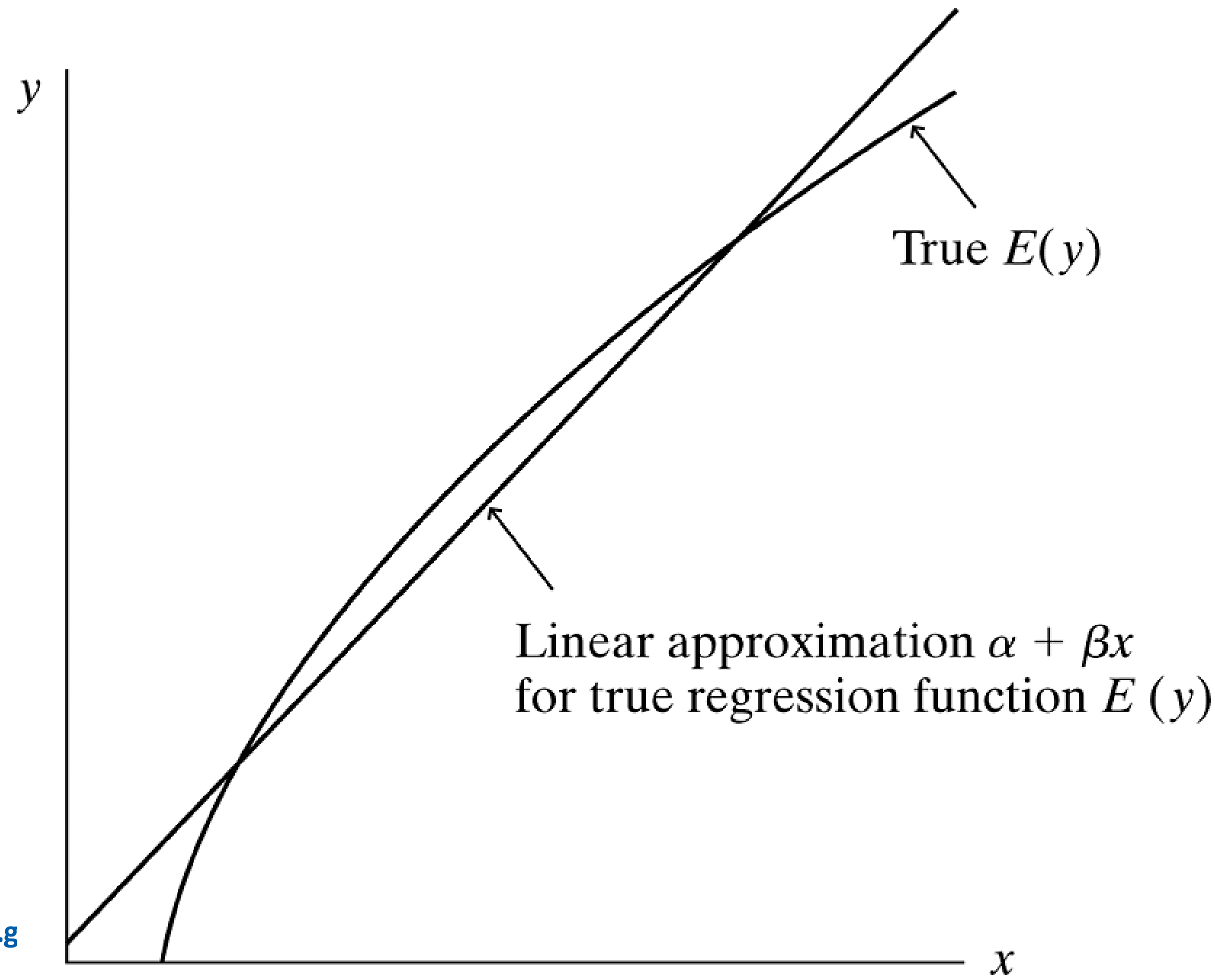
Model Assumptions and Violations

- We end this chapter by reconsidering the assumptions underlying linear regression analysis.
- We discuss the effects of violating these assumptions and the effects of *influential* observations.
- Finally, we show an alternate way to express the model.

Which Assumptions are Important?

- The linear regression model assumes that the relationship between x and the mean of y follows a straight line.
- The inferences introduced in the previous section are appropriate for detecting positive or negative linear associations.

Which Assumptions are Important?



A Linear Regression Equation as an Approximation for a Nonlinear Relationship

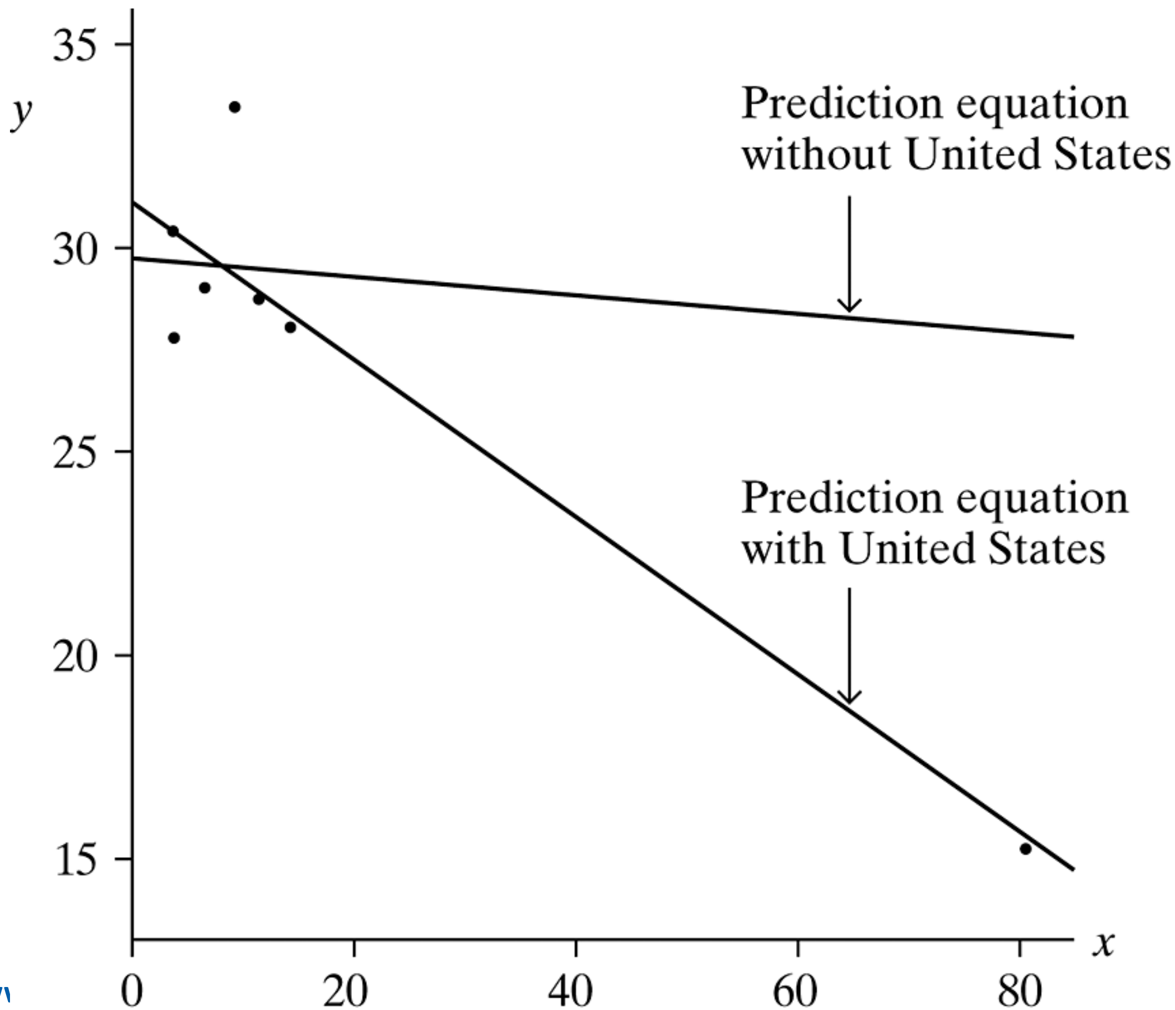
Which Assumptions are Important?

- The least squares line and r and r^2 are valid descriptive statistics no matter what the shape of the conditional distribution of y-values for each x-value.
- The random sample and straight-line assumptions are very important.

Influential Observations

- A single observation can have a large effect if it is a *regression outlier*—having *x*-value relatively large or relatively small and falling quite far from the trend that the rest of the data follow.
- The figure on the next slide illustrates this. The figure plots observations for several African and Asian nations on y = crude birth rate (number of births per 1000 population size) and x = number of televisions per 100 people.

Influential Observations



Prediction Equations for y = Birth Rate and x = Television Ownership, with and without Observation for the United States

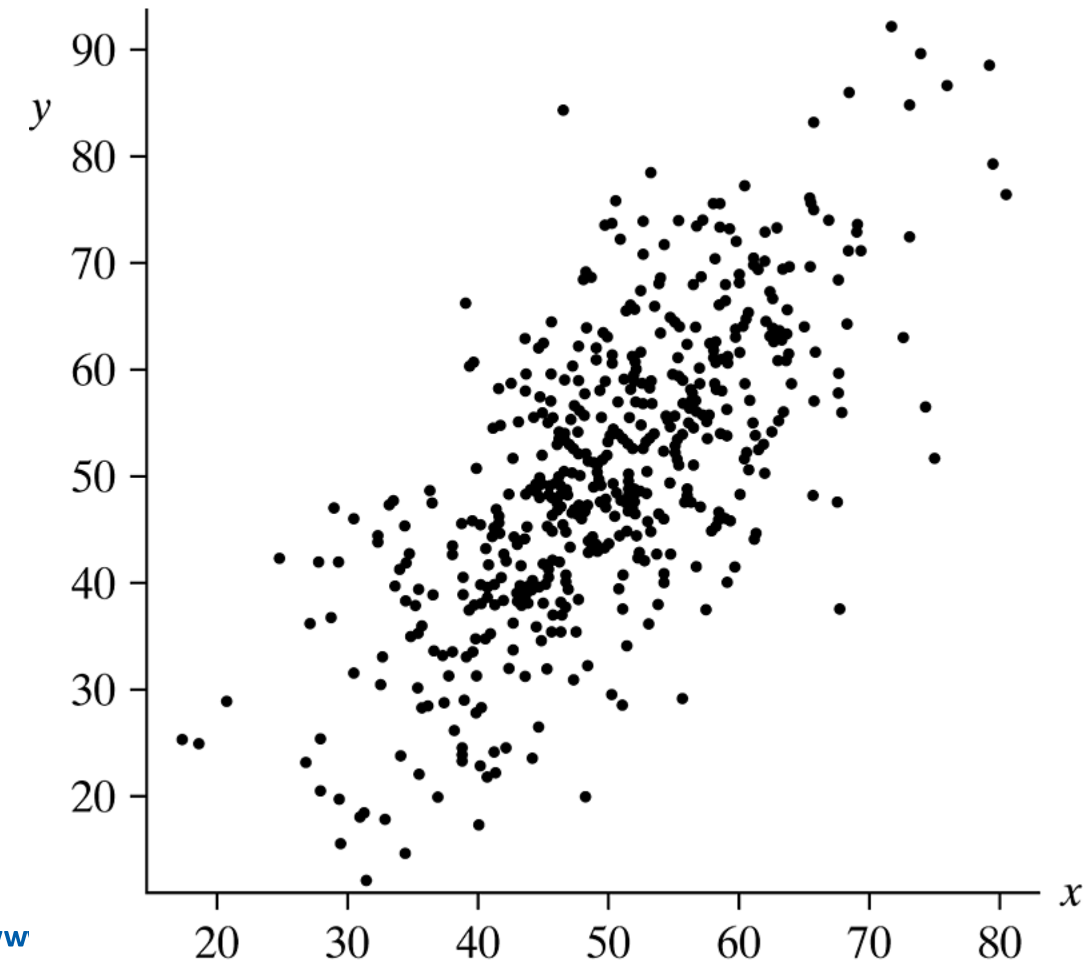
Influential Observations

- When a scatterplot shows a severe regression outlier, you should investigate the reasons for it.
- Observations that have a large influence on the model parameter estimates can also have a large impact on the correlation.

Factors Influencing the Correlation

- When a sample has a much narrower range of variation in x than the population, the sample correlation tends to underestimate drastically (in absolute value) the population correlation.
- The correlation is most appropriate as a summary measure of association when the sample (x, y)-values are a random sample of the population.

Factors Influencing the Correlation



The Correlation Is Affected by the Range of *x*-Values. The correlation decreases from 0.71 to 0.33 using only points with *x* between 43 and 57.

Example: Does the SAT Predict College GPA?

Consider the association between *x* = score on the SAT college entrance exam and *y* = college GPA at end of second year of college. The strength of the correlation depends on the variability in SAT scores in the sample. If we study the association only for students at Harvard University, the correlation will probably be weak, because the sample SAT scores will be concentrated very narrowly at the upper end of the scale. By contrast, if we could randomly sample from the population of *all* high school students who take the SAT and then place those students in the Harvard environment, students with poor SAT scores would tend to have low GPAs at Harvard. We would then observe a much stronger correlation.

Extrapolation is Dangerous

- It is dangerous to apply a prediction equation to values of *x* outside the range of observed values.
- Here is another type of inappropriate extrapolation: *x* being positively correlated with *y* and *y* being positively correlated with *z* does not imply that *x* is positively correlated with *z*.

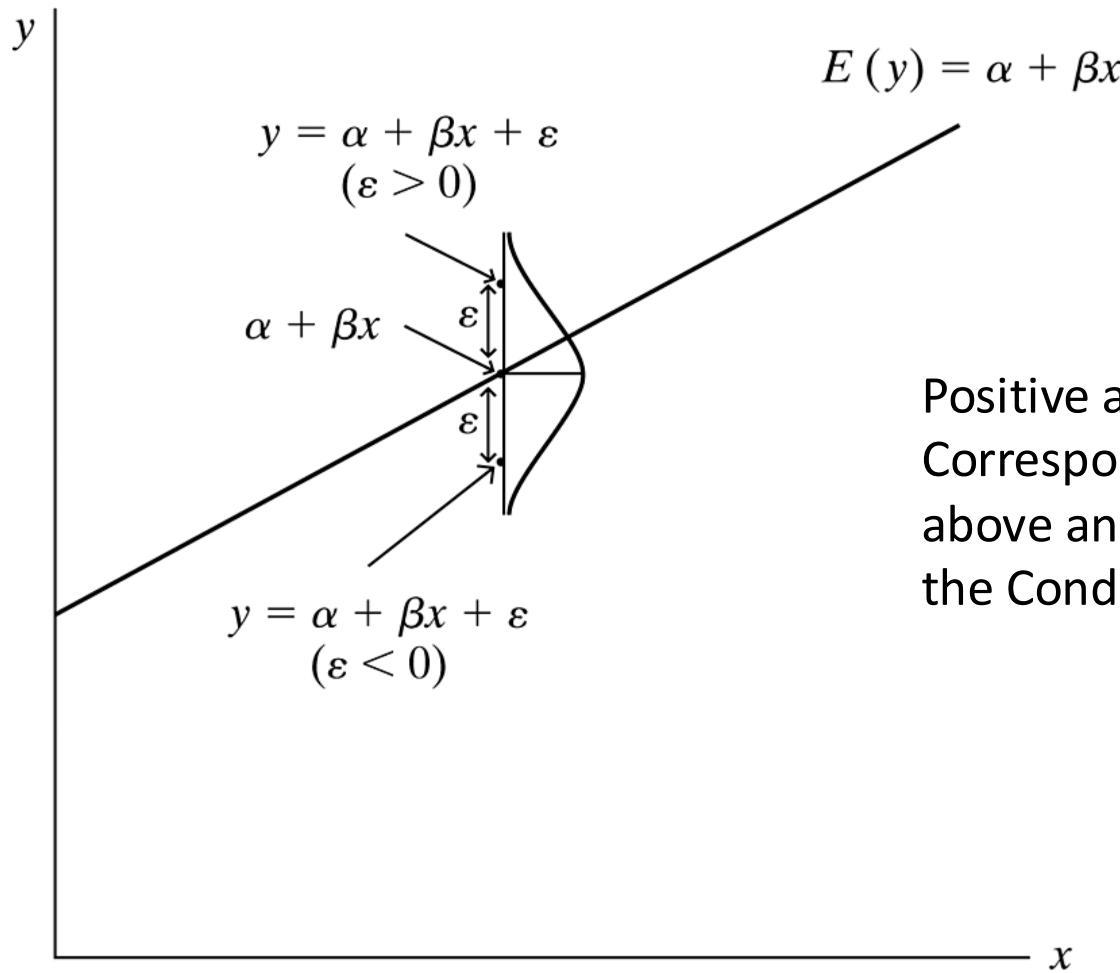
Regression Model with Error Terms

- We've seen that the deterministic model $y = \alpha + \beta x$ is unrealistic, because of not allowing variability of y-values. To allow variability, we include a term for the deviation of the observation y from the mean,

$$y = \alpha + \beta x + \varepsilon.$$

- The term denoted by ε (the Greek letter epsilon) represents the deviation of y from the mean, $\alpha + \beta x$. Each observation has its own value for ε.

Regression Model with Error Terms



Positive and Negative ε-Values Correspond to Observations above and below the Mean of the Conditional Distribution

Regression Model with Error Terms

- For each x, variability in the y-values corresponds to variability in ε. The ε term is called the ***error term***, since it represents the error that results from using the mean value ($\alpha + \beta x$) of y at a certain value of x to predict the individual observation.

Regression Model with Error Terms

- For the sample data and their prediction equation, let e be such that

$$y = a + bx + e.$$

- That is, $y = \hat{y} + e$, so $e = y - \hat{y}$. Then e is the *residual*, the difference between the observed and predicted values of y, which we *can* observe. Since $y = \alpha + \beta x + \varepsilon$, the residual e estimates ε. We can interpret ε as a ***population residual***.

Model Reality, and Alternative Approaches

- The regression model *approximates* the true relationship.
- If the model seems too simple to be adequate, the scatterplot or other diagnostics may suggest improvement by using other models introduced later in this text.

Summary

This lecture dealt with *association between quantitative variables*. A new element studied here was a regression model to describe the *form* of the relationship between the explanatory variable x and the mean $E(y)$ of the response variable. The major aspects of the analysis are as follows:

Summary

- The ***linear regression equation*** $E(y) = \alpha + \beta x$ describes the *form* of the relationship. This equation is appropriate when a straight line approximates the relationship between x and the mean of y.
- A ***scatterplot*** views the data and checks whether the relationship is approximately linear. If it is, the ***least squares*** estimates of the y-intercept α and the slope β provide the prediction equation $\hat{y} = a + bx$ closest to the data, minimizing the sum of squared residuals.

Summary

- The ***correlation r*** and its square describe the *strength* of the linear association. The correlation is a standardized slope, having the same sign as the slope but falling between −1 and +1. Its square, r^2, gives the proportional reduction in variability about the prediction equation compared to the variability about $\bar{y}$.
- For inference about the relationship, a *t* test using the slope or correlation tests the ***null hypothesis of independence***. A confidence interval estimates the size of the effect.

Chapter Summary

TABLE 9.9: Summary of Tests of Independence and Measures of Association

	Measurement Levels Of Variables		
	Nominal	Ordinal	Interval
Null hypothesis	H_0: independence	H_0: independence	H_0: independence ($\beta = 0$)
Test statistic	$X^2 = \sum \frac{(f_0 - f_e)^2}{f_e}$	$z = \frac{\hat{\gamma}}{se}$	$t = \frac{b}{se} = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$, $df = n - 2$
Measure of association	$\hat{\pi}_2 - \hat{\pi}_1$	$\hat{\gamma} = \frac{C-D}{C+D}$	$r = b\left(\frac{s_x}{s_y}\right)$
	Odds ratio		$r^2 = \frac{TSS-SSE}{TSS}$